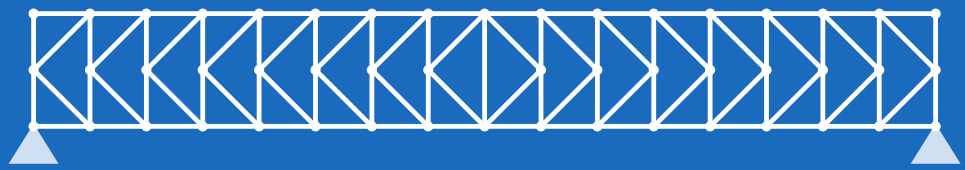
TRUSS REPORT

K-truss bridge 64 m (rigid joints)

Plane truss — analysis and CIRSOC 301 rod checks



Author A. Student · Course Structures I · 13/10/2026

MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+943.8 kN	-944.0 kN	84.0 mm	0.75	8.66	15679 kg
rod 8	rod 24	L/762 · limit L/300	rod 8 · H1-1a, axial + bending	the load can grow ×8.7	carries 900.0 kN · 6× own weight

The model

Span	64.00 m
Height	8.00 m
Nodes / rods	50 / 97
Supports	pin at 0, rollerX at 16
Loads	15 nodal, total 900.0 kN
Rod families	5, with their sections:
Bottom chord	HEA 220, F24 (CIRSOC) / A36
Diagonal	SHS 150x150x6, F24 (CIRSOC) / A36
End post	SHS 200x200x8, F24 (CIRSOC) / A36
Top chord	HEB 220, F24 (CIRSOC) / A36
Vertical	SHS 120x120x5, F24 (CIRSOC) / A36

What the analysis found

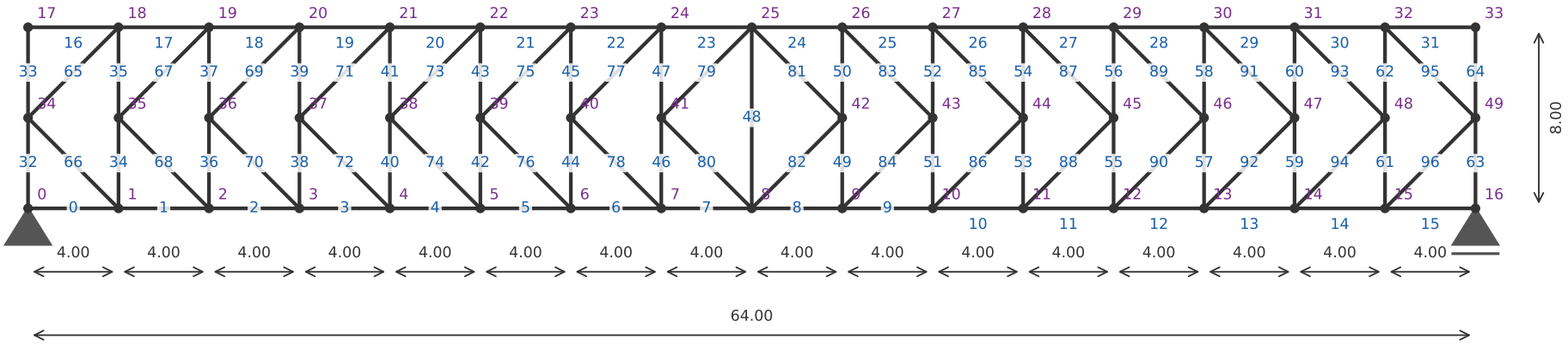
Stability	Statically indeterminate to degree 144 (deformations + r − DOF = 291 + 3 − 150 = 144): the forces also depend on the stiffness of the rods.
Most used rod	rod 8 at 0.75 (H1-1a, axial + bending)
Rods over capacity	0 of 97 checked
Buckling	$\lambda_1 = 8.66$
Shear and moment	present — see the Shear, Moment, Joints and Rod detail sheets
Joint moment	largest -4.94 kN·m at joint 0 (clockwise, from rod 0)

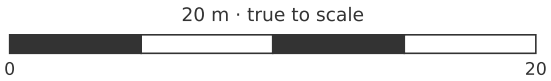
Contents

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Shear V
9 Moment M
10 Joints · 3 sheets
13 Node force vectors · 7 sheets
20 Rods · 2 sheets
22 Rod calculations · 7 sheets
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35 Explain rod 8 and 7

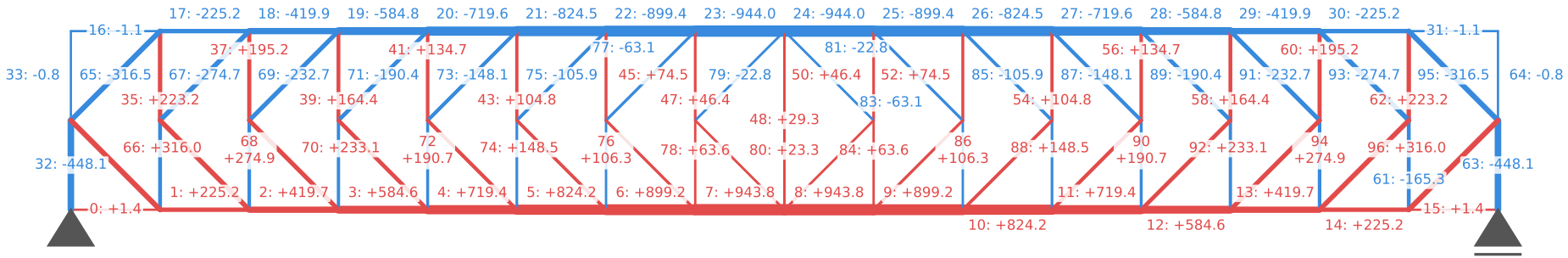
Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.





Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 944.0).

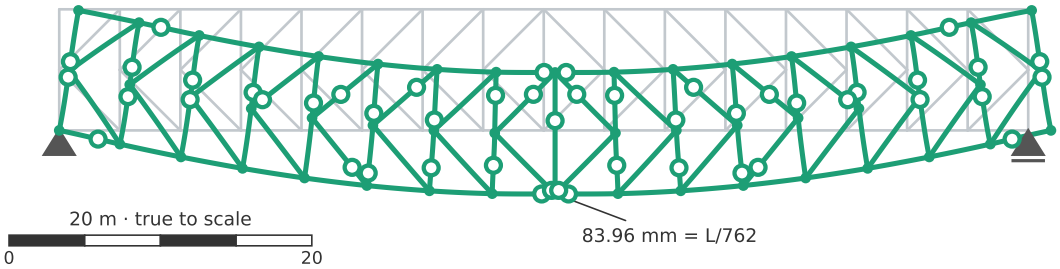


20 m · true to scale

0 20

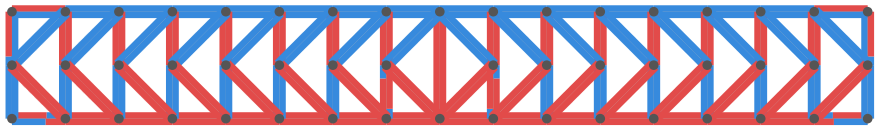
14 of 97 labels left out where they would overlap; every force is in the Rods table.

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. ○ = point of contraflexure (M = 0), where a rod changes the way it bends: 49 of them. Largest vertical deflection 83.96 mm against the limit L/300 = 213.3 mm: OK.

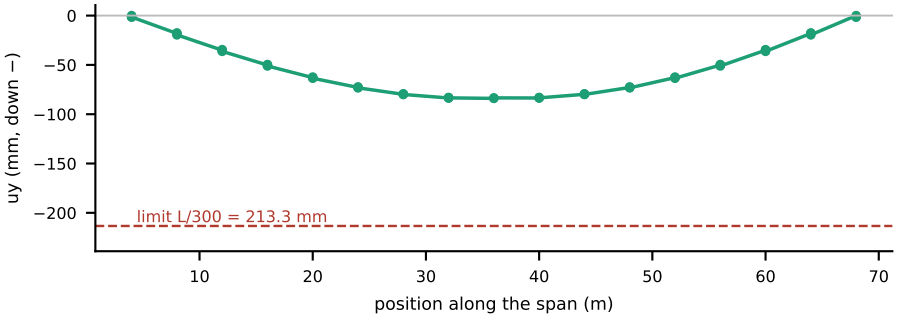


Fibre stress top and bottom face of each member

red = tension, blue = compression. Axial force and bending together (N/A ± M·c/I); a band flips colour at a point of contraflexure.

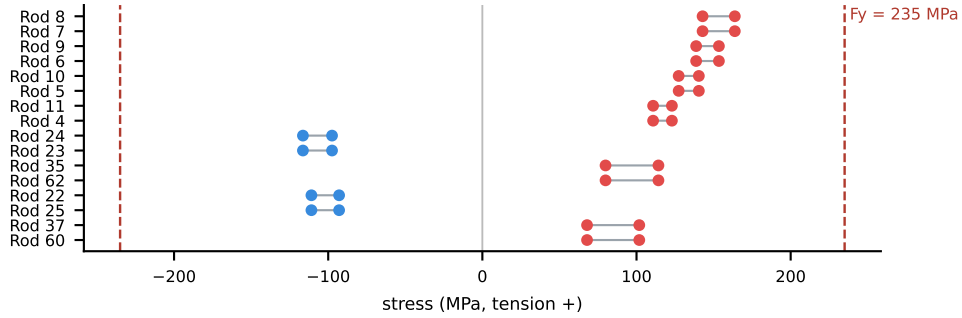


Deflection along the span vertical movement of every joint



Largest 83.96 mm at joint 8 = L/762. Within the limit.

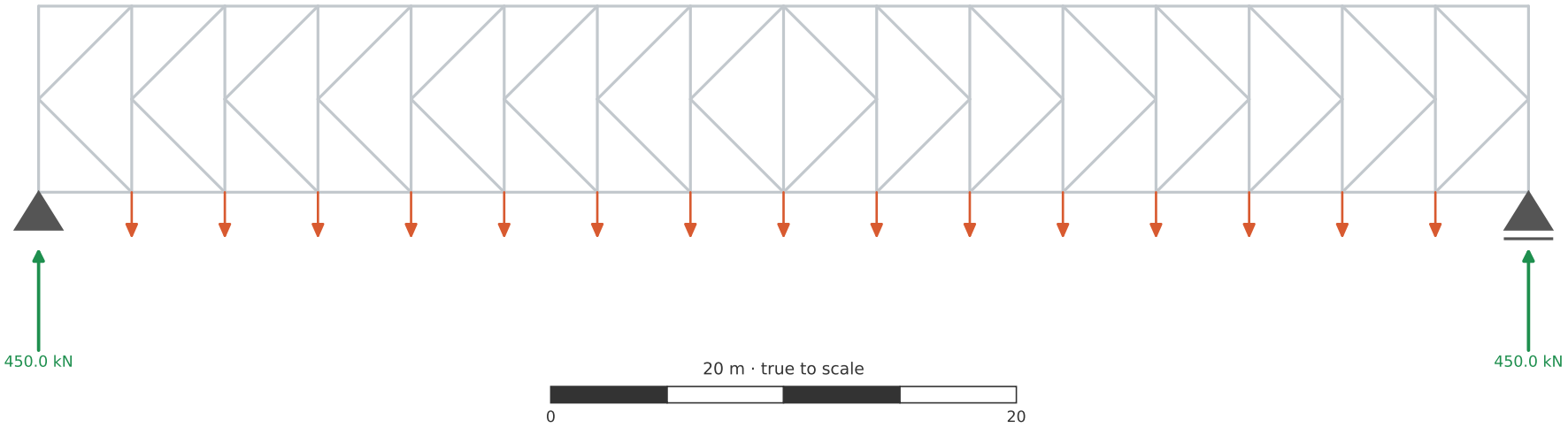
Fibre stress values the two faces of each rod, MPa



The 16 most stressed rods; the bar between a rod's two dots is its bending.

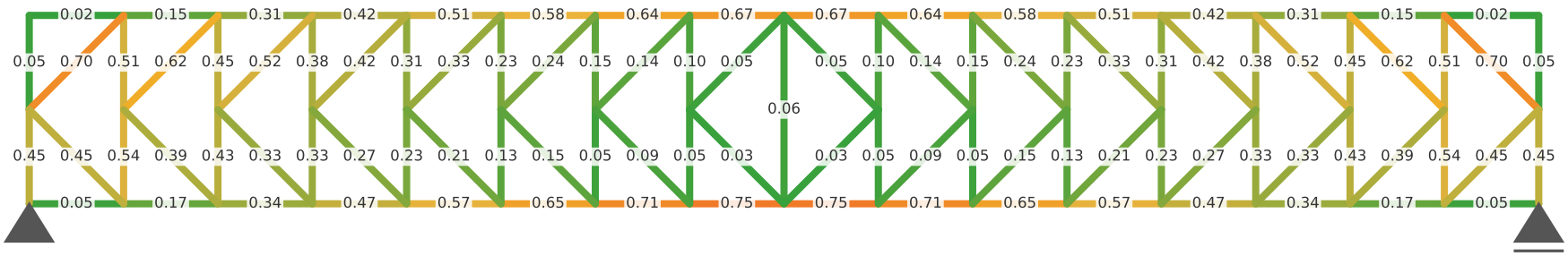
Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

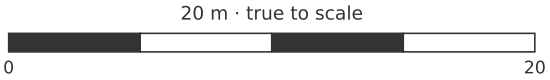
15 loads of 60.0 kN each, hung from the joints

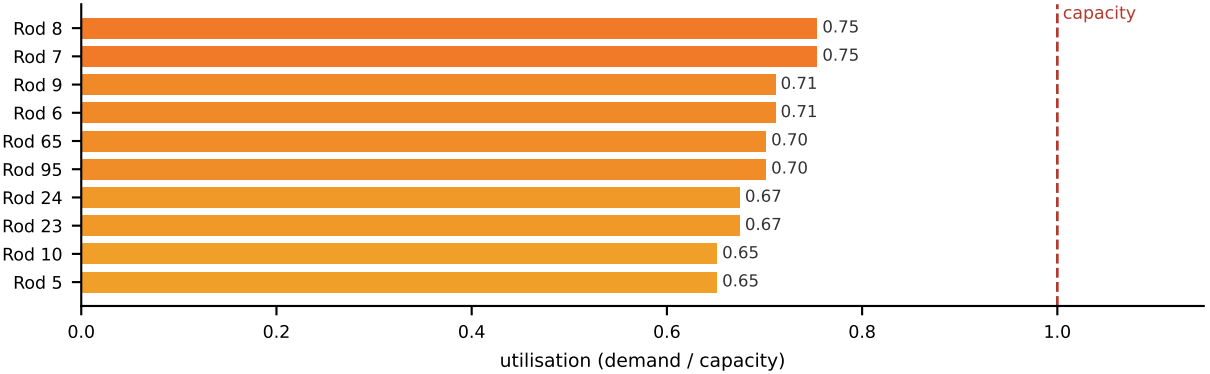


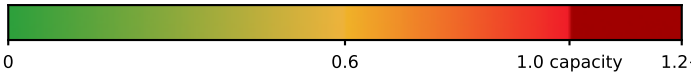
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	450.00
16	rollerX	0.00	450.00
Σ		0.00	900.00

Utilisation: demand over CIRSOC 301 capacity. Most used rod 8 at 0.75 (H1-1a, axial + bending). Every checked rod is within capacity.









grey = no steel section, not checked

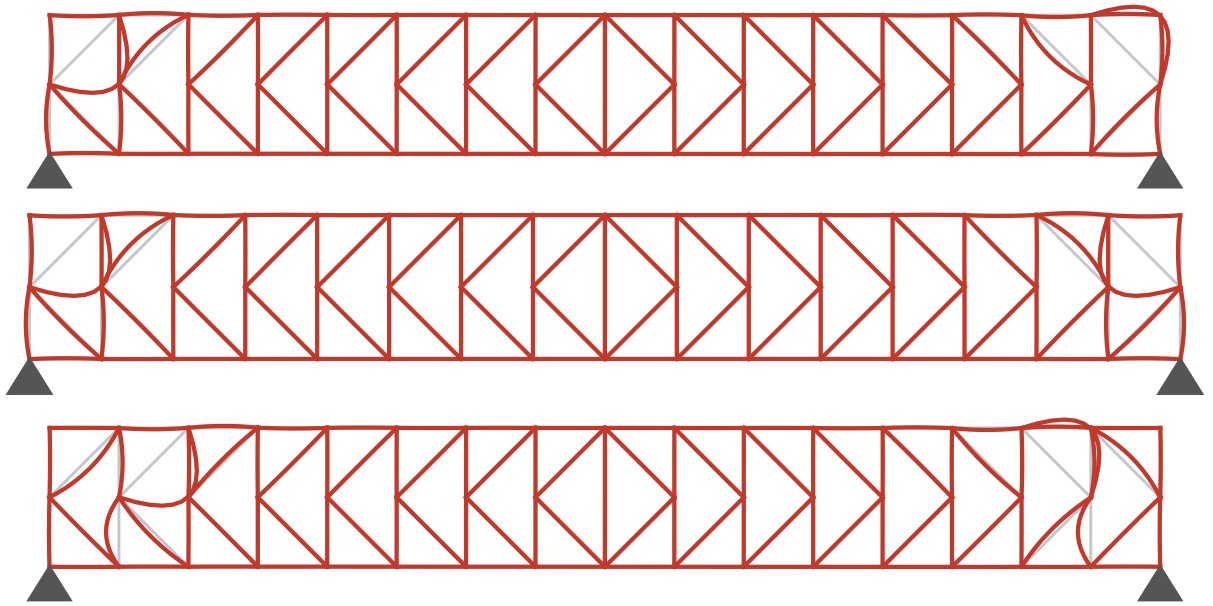
The 10 most used of 97 checked rods; the rest are in the Rods table.

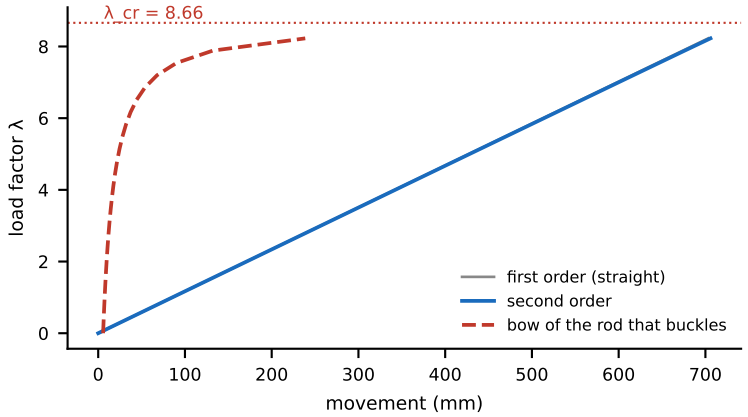
λ is the factor on the load as it stands. $\lambda_1 = 8.66$, so the load can grow 8.7× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

Mode 1: λ = 8.66
rod 95 bows

Mode 2: λ = 8.66
rod 65 bows

Mode 3: λ = 9.00
rod 67 bows

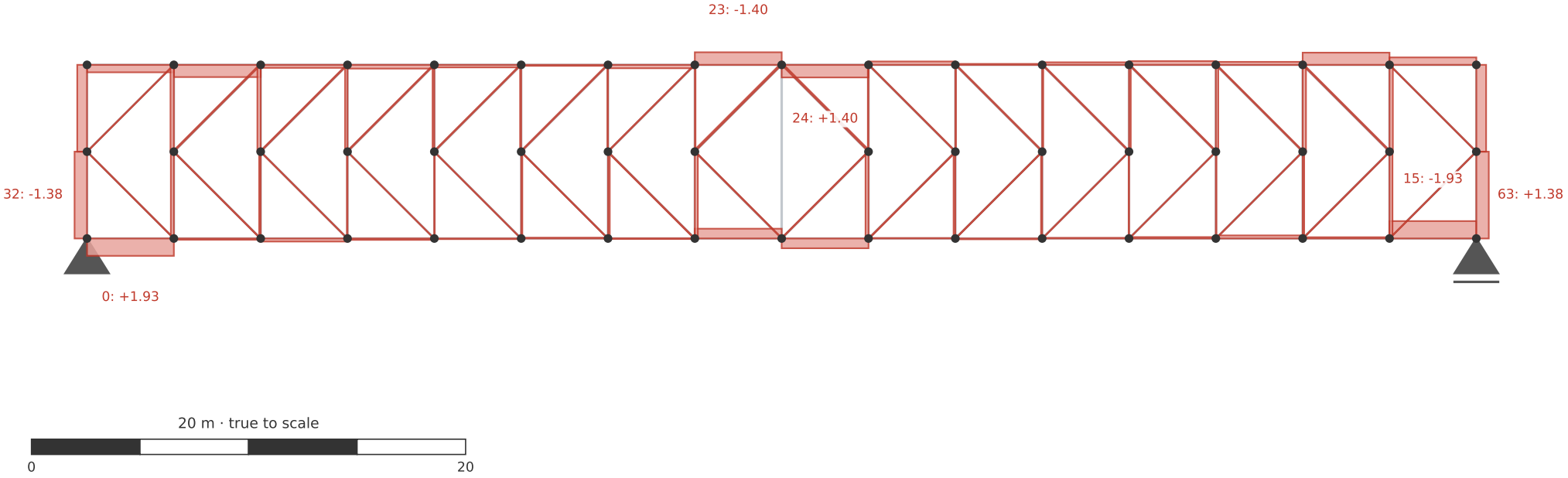




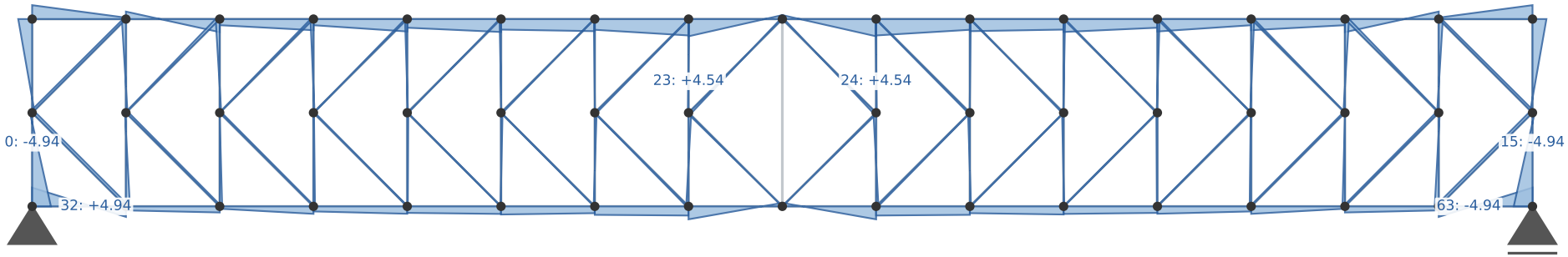
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.66 mm in the first mode's shape.

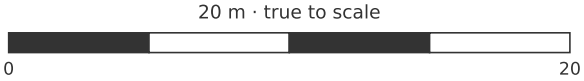
Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Shear force V along every rigid rod. Biggest +1.93 kN in rod 0; the band is drawn off each member, proportional to the value, and labelled on the six largest.

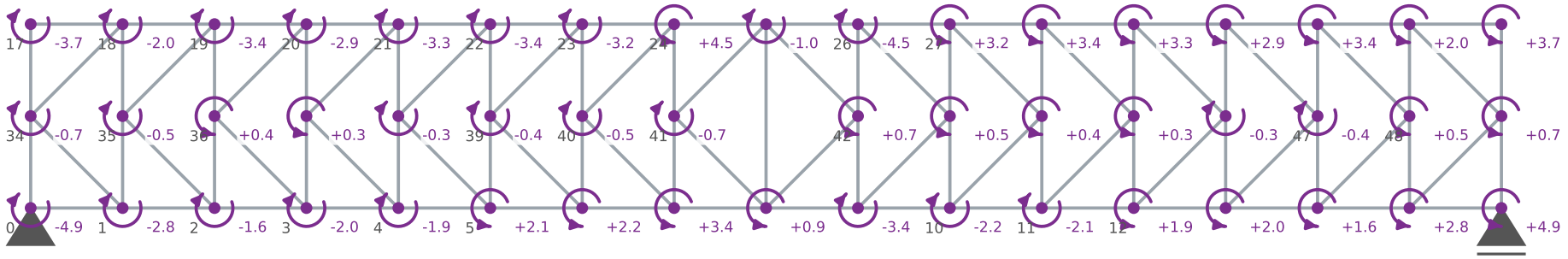


Bending moment M along every rigid rod, drawn on the face it puts in TENSION, as on screen (sagging positive: below a horizontal rod). Biggest -4.94 kN·m in rod 0; the band is drawn off each member, proportional to the value, and labelled on the six largest.





The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

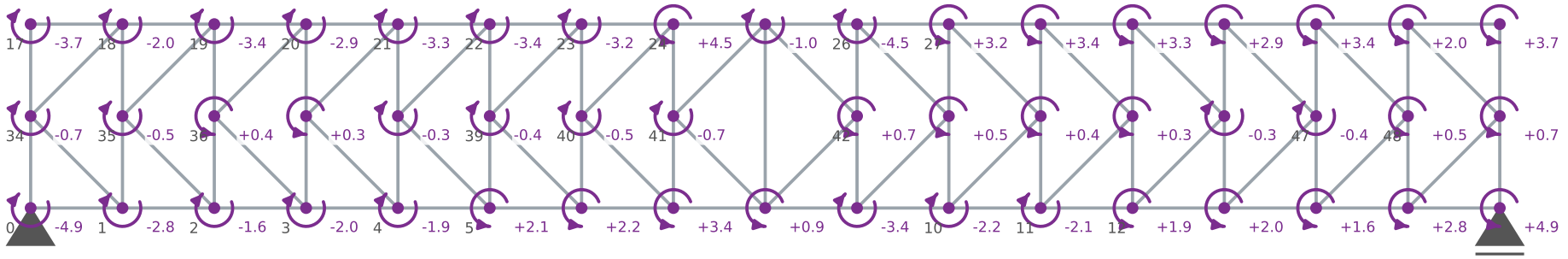
0 20

22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
0	rigid · pin	4.94	−	clockwise	0	top	0.00 ✓
16	rigid · rollerX	4.94	+	anticlockwise	15	top	0.00 ✓
26	rigid	4.54	−	clockwise	24	bottom	0.00 ✓
24	rigid	4.54	+	anticlockwise	23	bottom	0.00 ✓
33	rigid	3.67	+	anticlockwise	31	top	0.00 ✓
17	rigid	3.67	−	clockwise	16	top	0.00 ✓
31	rigid	3.45	+	anticlockwise	30	bottom	0.00 ✓
19	rigid	3.45	−	clockwise	17	bottom	0.00 ✓
9	rigid	3.41	−	clockwise	8	bottom	0.00 ✓
7	rigid	3.41	+	anticlockwise	7	bottom	0.00 ✓
28	rigid	3.40	+	anticlockwise	27	bottom	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
22	rigid	3.40	−	clockwise	20	bottom	0.00 ✓
21	rigid	3.27	−	clockwise	19	bottom	0.00 ✓
29	rigid	3.27	+	anticlockwise	28	bottom	0.00 ✓
27	rigid	3.19	+	anticlockwise	26	bottom	0.00 ✓
23	rigid	3.19	−	clockwise	21	bottom	0.00 ✓
20	rigid	2.94	−	clockwise	18	bottom	0.00 ✓
30	rigid	2.94	+	anticlockwise	29	bottom	0.00 ✓
1	rigid	2.77	−	clockwise	0	bottom	0.00 ✓
15	rigid	2.77	+	anticlockwise	15	bottom	0.00 ✓
6	rigid	2.23	+	anticlockwise	6	bottom	0.00 ✓
10	rigid	2.23	−	clockwise	9	bottom	0.00 ✓

The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

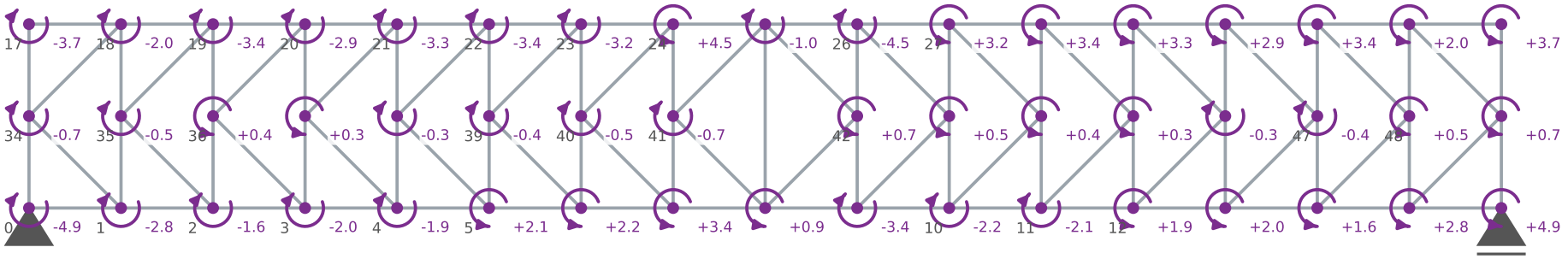
0 20

22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
11	rigid	2.14	−	clockwise	10	bottom	0.00 ✓
5	rigid	2.14	+	anticlockwise	5	bottom	0.00 ✓
32	rigid	1.97	+	anticlockwise	30	top	0.00 ✓
18	rigid	1.97	−	clockwise	17	top	0.00 ✓
3	rigid	1.97	−	clockwise	2	bottom	0.00 ✓
13	rigid	1.97	+	anticlockwise	13	bottom	0.00 ✓
12	rigid	1.94	+	anticlockwise	12	bottom	0.00 ✓
4	rigid	1.94	−	clockwise	3	bottom	0.00 ✓
14	rigid	1.60	+	anticlockwise	14	bottom	0.00 ✓
2	rigid	1.60	−	clockwise	1	bottom	0.00 ✓
25	rigid	1.04	−	clockwise	24	top	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
8	rigid	0.93	+	anticlockwise	7	top	0.00 ✓
42	rigid	0.70	+	anticlockwise	49	left	0.00 ✓
41	rigid	0.70	−	clockwise	46	right	0.00 ✓
34	rigid	0.67	−	clockwise	66	top	0.00 ✓
49	rigid	0.67	+	anticlockwise	96	top	0.00 ✓
48	rigid	0.53	+	anticlockwise	94	top	0.00 ✓
35	rigid	0.53	−	clockwise	68	top	0.00 ✓
43	rigid	0.48	+	anticlockwise	51	left	0.00 ✓
40	rigid	0.48	−	clockwise	44	right	0.00 ✓
36	rigid	0.41	+	anticlockwise	37	right	0.00 ✓
47	rigid	0.41	−	clockwise	60	left	0.00 ✓

The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

0 20

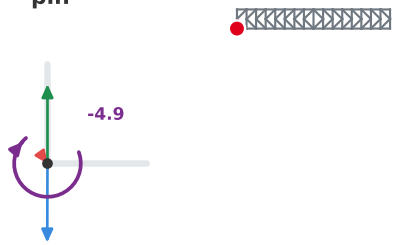
22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
39	rigid	0.40	−	clockwise	42	right	0.00 ✓
44	rigid	0.40	+	anticlockwise	53	left	0.00 ✓
46	rigid	0.28	−	clockwise	58	left	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
37	rigid	0.28	+	anticlockwise	39	right	0.00 ✓
38	rigid	0.27	−	clockwise	40	right	0.00 ✓
45	rigid	0.27	+	anticlockwise	55	left	0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

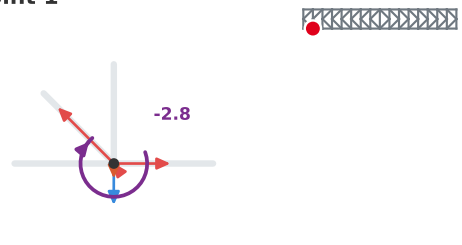
Joint 0 · pin



member	N	θ°	Fx	Fy	M
r0→1	+1.4	305.6	+1.4	-1.9	-4.9
r32→34	-448.1	269.8	-1.4	-448.1	+4.9
reaction			+0.0	+450.0	

max M -4.94 kN·m rod 0 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

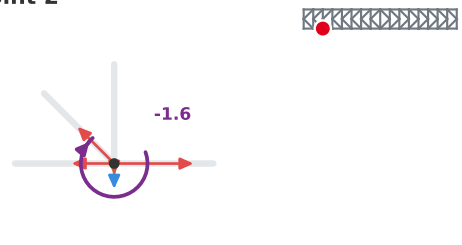
Joint 1



member	N	θ°	Fx	Fy	M
r0→0	+1.4	125.6	-1.4	+1.9	-2.8
r1→2	+225.2	360.0	+225.2	-0.1	+1.1
r34→35	-165.3	269.9	-0.3	-165.3	+1.1
r66→34	+316.0	135.0	-223.5	+223.5	+0.7
load			+0.0	-60.0	

max M -2.77 kN·m rod 0 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

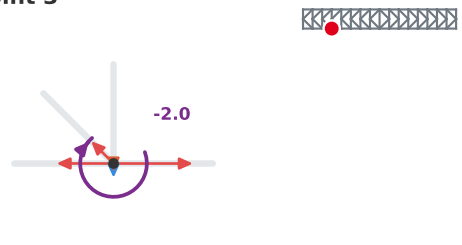
Joint 2



member	N	θ°	Fx	Fy	M
r1→1	+225.2	180.0	-225.2	+0.1	-1.6
r2→3	+419.7	360.0	+419.7	-0.3	+0.6
r36→36	-134.2	269.9	-0.2	-134.2	+0.6
r68→35	+274.9	135.0	-194.3	+194.4	+0.3
load			+0.0	-60.0	

max M -1.60 kN·m rod 1 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

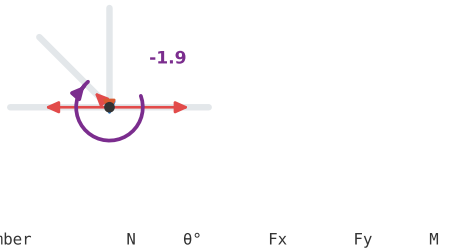
Joint 3



member	N	θ°	Fx	Fy	M
r2→2	+419.7	180.0	-419.7	+0.3	-2.0
r3→4	+584.6	360.0	+584.6	-0.2	+1.3
r38→37	-105.0	270.0	-0.1	-105.0	+0.4
r70→36	+233.1	135.0	-164.8	+164.8	+0.2
load			+0.0	-60.0	

max M -1.97 kN·m rod 2 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

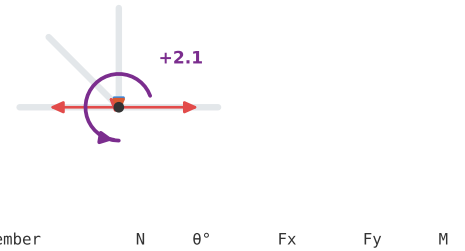
Joint 4



member	N	θ°	Fx	Fy	M
r3→3	+584.6	180.0	-584.6	+0.2	-1.9
r4→5	+719.4	360.0	+719.4	-0.1	+1.7
r40→38	-75.0	270.0	+0.0	-75.0	+0.1
r72→37	+190.7	135.0	-134.9	+134.9	+0.1
load			+0.0	-60.0	

max M -1.94 kN·m rod 3 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

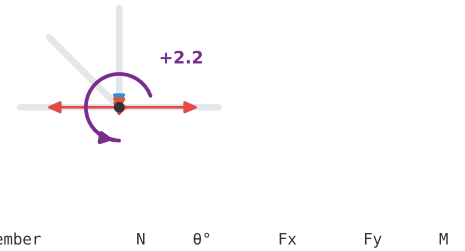
Joint 5



member	N	θ°	Fx	Fy	M
r4→4	+719.4	180.0	-719.4	+0.1	-2.0
r5→6	+824.2	0.0	+824.2	+0.1	+2.1
r42→39	-45.2	270.2	+0.1	-45.2	-0.1
r74→38	+148.5	135.0	-105.0	+105.0	-0.0
load			+0.0	-60.0	

max M +2.14 kN·m rod 5 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

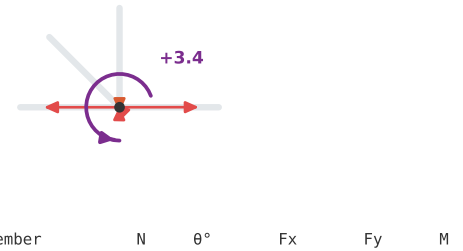
Joint 6



member	N	θ°	Fx	Fy	M
r5→5	+824.2	180.0	-824.2	-0.1	-1.8
r6→7	+899.2	360.0	+899.2	-0.0	+2.2
r44→40	-15.1	270.7	+0.2	-15.1	-0.3
r76→39	+106.3	135.0	-75.2	+75.2	-0.1
load			+0.0	-60.0	

max M +2.23 kN·m rod 6 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 7

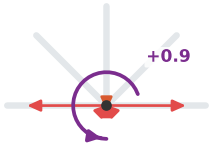


member	N	θ°	Fx	Fy	M
r6→6	+899.2	180.0	-899.2	+0.0	-2.4
r7→8	+943.8	0.1	+943.8	+1.1	+3.4
r46→41	+13.9	88.6	+0.3	+13.9	-0.6
r78→40	+63.6	134.9	-44.9	+45.0	-0.4
load			+0.0	-60.0	

max M +3.41 kN·m rod 7 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

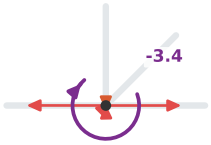
Joint 8



member	N	θ°	Fx	Fy	M
r7→7	+943.8	180.1	-943.8	-1.1	+0.9
r8→9	+943.8	359.9	+943.8	-1.1	-0.9
r48→25	+29.3	90.0	+0.0	+29.3	+0.0
r80→41	+23.3	135.1	-16.5	+16.5	+0.0
r82→42	+23.3	44.9	+16.5	+16.5	-0.0
load			+0.0	-60.0	

max M +0.93 kN·m rod 7 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

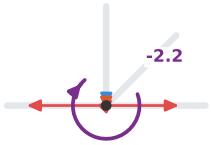
Joint 9



member	N	θ°	Fx	Fy	M
r8→8	+943.8	179.9	-943.8	+1.1	-3.4
r9→10	+899.2	0.0	+899.2	+0.0	+2.4
r49→42	+13.9	91.4	-0.3	+13.9	+0.6
r84→43	+63.6	45.1	+44.9	+45.0	+0.4
load			+0.0	-60.0	

max M -3.41 kN·m rod 8 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

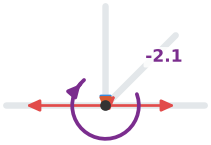
Joint 10



member	N	θ°	Fx	Fy	M
r9→9	+899.2	180.0	-899.2	-0.0	-2.2
r10→11	+824.2	360.0	+824.2	-0.1	+1.8
r51→43	-15.1	269.3	-0.2	-15.1	+0.3
r86→44	+106.3	45.0	+75.2	+75.2	+0.1
load			+0.0	-60.0	

max M -2.23 kN·m rod 9 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

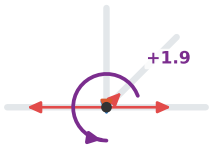
Joint 11



member	N	θ°	Fx	Fy	M
r10→10	+824.2	180.0	-824.2	+0.1	-2.1
r11→12	+719.4	0.0	+719.4	+0.1	+2.0
r53→44	-45.2	269.8	-0.1	-45.2	+0.1
r88→45	+148.5	45.0	+105.0	+105.0	+0.0
load			+0.0	-60.0	

max M -2.14 kN·m rod 10 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

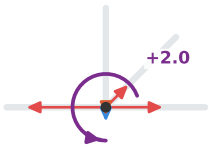
Joint 12



member	N	θ°	Fx	Fy	M
r11→11	+719.4	180.0	-719.4	-0.1	-1.7
r12→13	+584.6	0.0	+584.6	+0.2	+1.9
r55→45	-75.0	270.0	-0.0	-75.0	-0.1
r90→46	+190.7	45.0	+134.9	+134.9	-0.1
load			+0.0	-60.0	

max M +1.94 kN·m rod 12 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

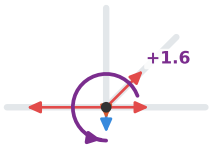
Joint 13



member	N	θ°	Fx	Fy	M
r12→12	+584.6	180.0	-584.6	-0.2	-1.3
r13→14	+419.7	0.0	+419.7	+0.3	+2.0
r57→46	-105.0	270.0	+0.1	-105.0	-0.4
r92→47	+233.1	45.0	+164.8	+164.8	-0.2
load			+0.0	-60.0	

max M +1.97 kN·m rod 13 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

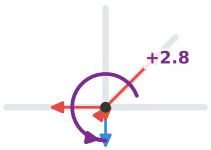
Joint 14



member	N	θ°	Fx	Fy	M
r13→13	+419.7	180.0	-419.7	-0.3	-0.6
r14→15	+225.2	0.0	+225.2	+0.1	+1.6
r59→47	-134.2	270.1	+0.2	-134.2	-0.6
r94→48	+274.9	45.0	+194.3	+194.4	-0.3
load			+0.0	-60.0	

max M +1.60 kN·m rod 14 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 15

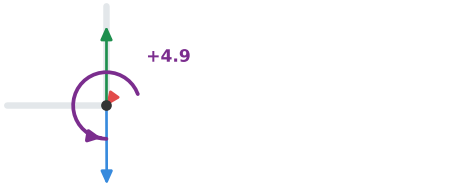


member	N	θ°	Fx	Fy	M
r14→14	+225.2	180.0	-225.2	-0.1	-1.1
r15→16	+1.4	54.4	+1.4	+1.9	+2.8
r61→48	-165.3	270.1	+0.3	-165.3	-1.1
r96→49	+316.0	45.0	+223.5	+223.5	-0.7
load			+0.0	-60.0	

max M +2.77 kN·m rod 15 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

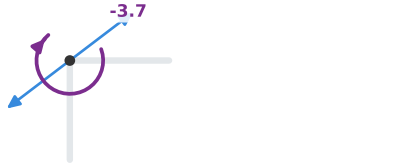
Joint 16 · rollerX



member	N	θ°	Fx	Fy	M
r15→15	+1.4	234.4	-1.4	-1.9	+4.9
r63→49	-448.1	270.2	+1.4	-448.1	-4.9
reaction			+0.0	+450.0	

max M +4.94 kN·m rod 15 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 17



member	N	θ°	Fx	Fy	M
r16→18	-1.1	217.3	-1.1	-0.8	-3.7
r33→34	-0.8	37.3	+1.1	+0.8	+3.7

max M -3.67 kN·m rod 16 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 18



member	N	θ°	Fx	Fy	M
r16→17	-1.1	37.3	+1.1	+0.8	+0.4
r17→19	-225.2	180.3	-225.2	-1.4	-2.0
r35→35	+223.2	270.1	+0.4	-223.2	+1.1
r65→34	-316.5	45.0	+223.8	+223.8	+0.5

max M -1.97 kN·m rod 17 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

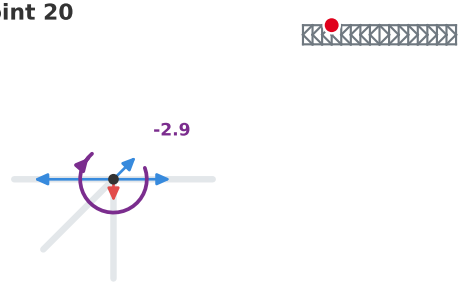
Joint 19



member	N	θ°	Fx	Fy	M
r17→18	-225.2	0.3	+225.2	+1.4	-3.4
r18→20	-419.9	180.0	-419.9	-0.3	+1.6
r37→36	+195.2	270.1	+0.4	-195.2	+1.1
r67→35	-274.7	45.0	+194.3	+194.1	+0.8

max M -3.45 kN·m rod 17 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

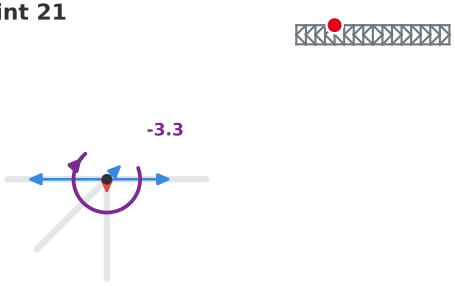
Joint 20



member	N	θ°	Fx	Fy	M
r18→19	-419.9	0.0	+419.9	+0.3	-2.9
r19→21	-584.8	180.0	-584.8	-0.4	+1.6
r39→37	+164.4	270.1	+0.3	-164.4	+0.8
r69→36	-232.7	45.0	+164.6	+164.5	+0.5

max M -2.94 kN·m rod 18 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

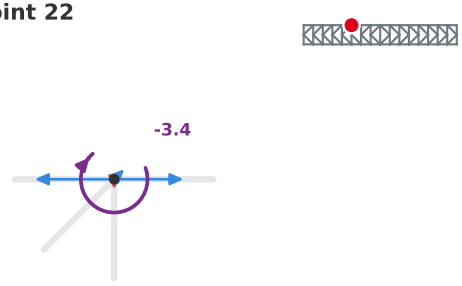
Joint 21



member	N	θ°	Fx	Fy	M
r19→20	-584.8	0.0	+584.8	+0.4	-3.3
r20→22	-719.6	180.0	-719.6	-0.3	+2.3
r41→38	+134.7	270.1	+0.2	-134.7	+0.6
r71→37	-190.4	45.0	+134.7	+134.5	+0.4

max M -3.27 kN·m rod 19 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

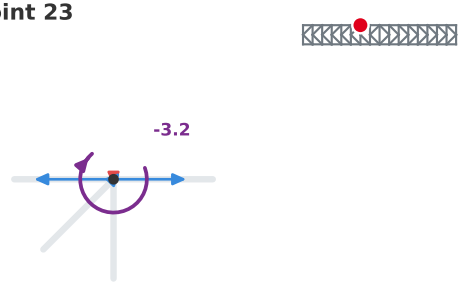
Joint 22



member	N	θ°	Fx	Fy	M
r20→21	-719.6	0.0	+719.6	+0.3	-3.4
r21→23	-824.5	180.0	-824.5	-0.1	+2.8
r43→39	+104.8	270.1	+0.1	-104.8	+0.4
r73→38	-148.1	45.0	+104.7	+104.6	+0.3

max M -3.40 kN·m rod 20 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 23

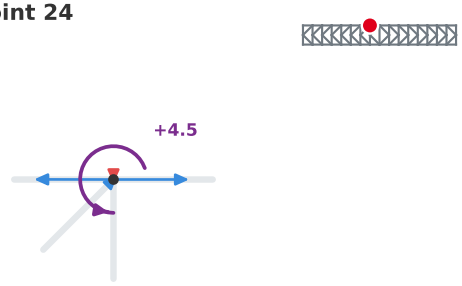


member	N	θ°	Fx	Fy	M
r21→22	-824.5	0.0	+824.5	+0.1	-3.2
r22→24	-899.4	180.0	-899.4	-0.4	+2.8
r45→40	+74.5	270.1	+0.1	-74.5	+0.2
r75→39	-105.9	45.0	+74.9	+74.8	+0.1

max M -3.19 kN·m rod 21 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

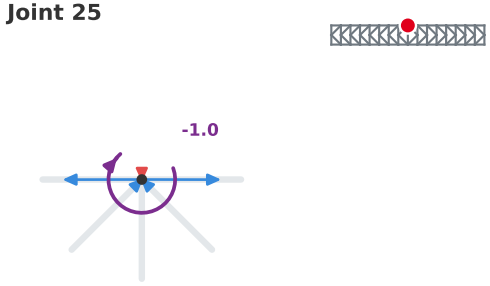
Joint 24



member	N	θ°	Fx	Fy	M
r22→23	-899.4	0.0	+899.4	+0.4	-4.3
r23→25	-944.0	179.9	-944.0	+1.4	+4.5
r47→41	+46.4	269.9	-0.1	-46.4	-0.1
r77→40	-63.1	45.0	+44.7	+44.6	-0.1

max M +4.54 kN·m rod 23 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

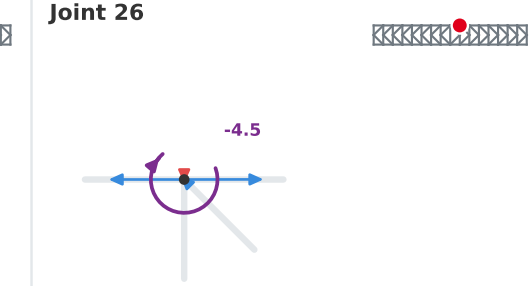
Joint 25



member	N	θ°	Fx	Fy	M
r23→24	-944.0	359.9	+944.0	-1.4	+1.0
r24→26	-944.0	180.1	-944.0	-1.4	-1.0
r48→8	+29.3	270.0	+0.0	-29.3	+0.0
r79→41	-22.8	44.6	+16.3	+16.0	+0.3
r81→42	-22.8	135.4	-16.3	+16.0	-0.3

max M -1.04 kN·m rod 24 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

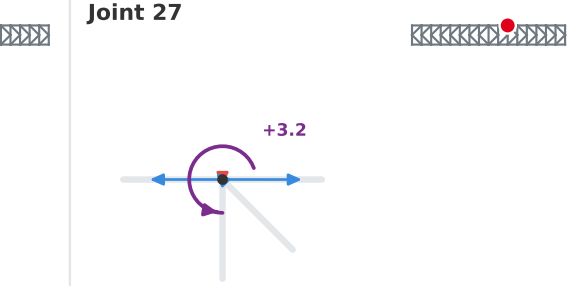
Joint 26



member	N	θ°	Fx	Fy	M
r24→25	-944.0	0.1	+944.0	+1.4	-4.5
r25→27	-899.4	180.0	-899.4	+0.4	+4.3
r50→42	+46.4	270.1	+0.1	-46.4	+0.1
r83→43	-63.1	135.0	-44.7	+44.6	+0.1

max M -4.54 kN·m rod 24 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

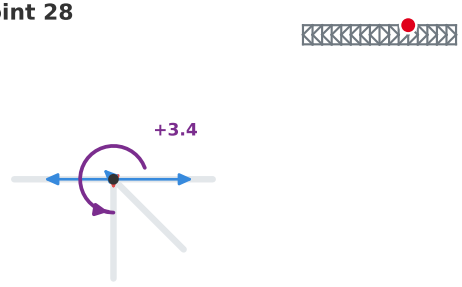
Joint 27



member	N	θ°	Fx	Fy	M
r25→26	-899.4	360.0	+899.4	-0.4	-2.8
r26→28	-824.5	180.0	-824.5	+0.1	+3.2
r52→43	+74.5	269.9	-0.1	-74.5	-0.2
r85→44	-105.9	135.0	-74.9	+74.8	-0.1

max M +3.19 kN·m rod 26 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

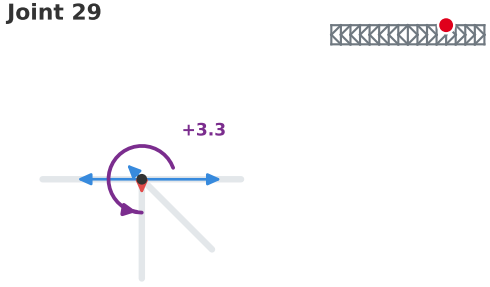
Joint 28



member	N	θ°	Fx	Fy	M
r26→27	-824.5	360.0	+824.5	-0.1	-2.8
r27→29	-719.6	180.0	-719.6	+0.3	+3.4
r54→44	+104.8	269.9	-0.1	-104.8	-0.4
r87→45	-148.1	135.0	-104.7	+104.6	-0.3

max M +3.40 kN·m rod 27 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

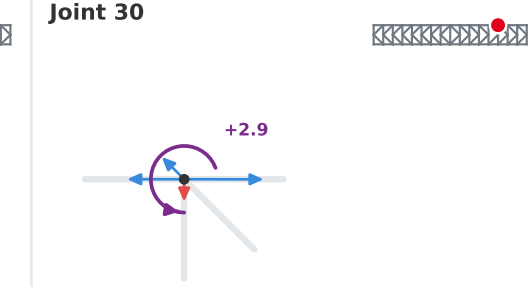
Joint 29



member	N	θ°	Fx	Fy	M
r27→28	-719.6	360.0	+719.6	-0.3	-2.3
r28→30	-584.8	180.0	-584.8	+0.4	+3.3
r56→45	+134.7	269.9	-0.2	-134.7	-0.6
r89→46	-190.4	135.0	-134.7	+134.5	-0.4

max M +3.27 kN·m rod 28 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

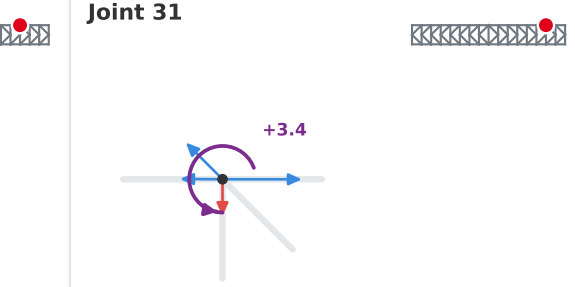
Joint 30



member	N	θ°	Fx	Fy	M
r28→29	-584.8	360.0	+584.8	-0.4	-1.6
r29→31	-419.9	180.0	-419.9	+0.3	+2.9
r58→46	+164.4	269.9	-0.3	-164.4	-0.8
r91→47	-232.7	135.0	-164.6	+164.5	-0.5

max M +2.94 kN·m rod 29 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 31

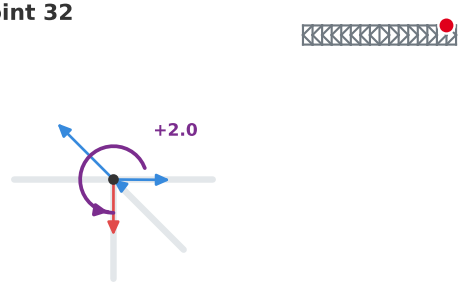


member	N	θ°	Fx	Fy	M
r29→30	-419.9	360.0	+419.9	-0.3	-1.6
r30→32	-225.2	179.7	-225.2	+1.4	+3.4
r60→47	+195.2	269.9	-0.4	-195.2	-1.1
r93→48	-274.7	135.0	-194.3	+194.1	-0.8

max M +3.45 kN·m rod 30 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

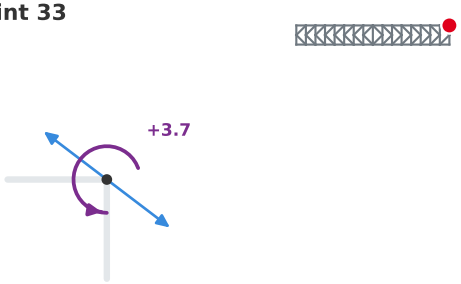
Joint 32



member	N	θ°	Fx	Fy	M
r30→31	-225.2	359.7	+225.2	-1.4	+2.0
r31→33	-1.1	142.7	-1.1	+0.8	-0.4
r62→48	+223.2	269.9	-0.4	-223.2	-1.1
r95→49	-316.5	135.0	-223.8	+223.8	-0.5

max M +1.97 kN·m rod 30 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

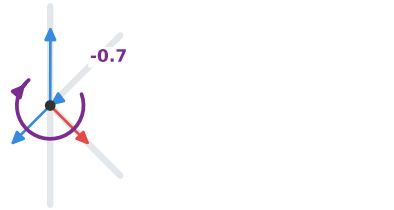
Joint 33



member	N	θ°	Fx	Fy	M
r31→32	-1.1	322.7	+1.1	-0.8	+3.7
r64→49	-0.8	142.7	-1.1	+0.8	-3.7

max M +3.67 kN·m rod 31 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 34



member	N	θ°	Fx	Fy	M
r32→0	-448.1	89.8	+1.4	+448.1	+0.6
r33→17	-0.8	217.3	-1.1	-0.8	+0.7
r65→18	-316.5	225.0	-223.8	-223.8	-0.6
r66→1	+316.0	315.0	+223.5	-223.5	-0.7

max M -0.67 kN·m rod 66 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

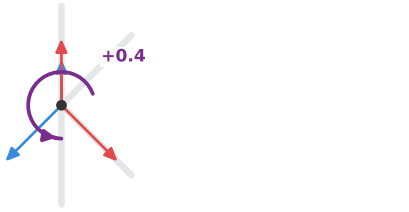
Joint 35



member	N	θ°	Fx	Fy	M
r34→1	-165.3	89.9	+0.3	+165.3	+0.3
r35→18	+223.2	90.1	-0.4	+223.2	+0.4
r67→19	-274.7	225.0	-194.3	-194.1	-0.1
r68→2	+274.9	315.0	+194.3	-194.4	-0.5

max M -0.53 kN·m rod 68 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 36



member	N	θ°	Fx	Fy	M
r36→2	-134.2	89.9	+0.2	+134.2	+0.0
r37→19	+195.2	90.1	-0.4	+195.2	+0.4
r69→20	-232.7	225.0	-164.6	-164.5	-0.1
r70→3	+233.1	315.0	+164.8	-164.8	-0.4

max M +0.41 kN·m rod 37 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 37



member	N	θ°	Fx	Fy	M
r38→3	-105.0	90.0	+0.1	+105.0	-0.1
r39→20	+164.4	90.1	-0.3	+164.4	+0.3
r71→21	-190.4	225.0	-134.7	-134.5	+0.1
r72→4	+190.7	315.0	+134.9	-134.9	-0.2

max M +0.28 kN·m rod 39 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 38



member	N	θ°	Fx	Fy	M
r40→4	-75.0	90.0	-0.0	+75.0	-0.3
r41→21	+134.7	90.1	-0.2	+134.7	+0.2
r73→22	-148.1	225.0	-104.7	-104.6	+0.2
r74→5	+148.5	315.0	+105.0	-105.0	-0.1

max M -0.27 kN·m rod 40 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 39



member	N	θ°	Fx	Fy	M
r42→5	-45.2	90.2	-0.1	+45.2	-0.4
r43→22	+104.8	90.1	-0.1	+104.8	+0.1
r75→23	-105.9	225.0	-74.9	-74.8	+0.3
r76→6	+106.3	315.0	+75.2	-75.2	+0.0

max M -0.40 kN·m rod 42 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 40



member	N	θ°	Fx	Fy	M
r44→6	-15.1	90.7	-0.2	+15.1	-0.5
r45→23	+74.5	90.1	-0.1	+74.5	+0.0
r77→24	-63.1	225.0	-44.7	-44.6	+0.4
r78→7	+63.6	314.9	+44.9	-45.0	+0.1

max M -0.48 kN·m rod 44 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

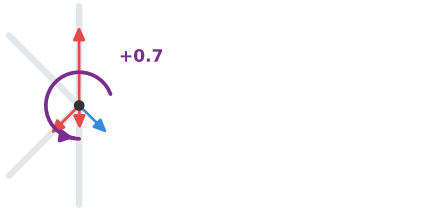
Joint 41



member	N	θ°	Fx	Fy	M
r46→7	+13.9	268.6	-0.3	-13.9	-0.7
r47→24	+46.4	89.9	+0.1	+46.4	-0.2
r79→25	-22.8	224.6	-16.3	-16.0	+0.6
r80→8	+23.3	315.1	+16.5	-16.5	+0.3

max M -0.70 kN·m rod 46 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

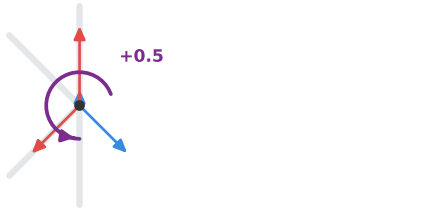
Joint 42



member	N	θ°	Fx	Fy	M
r49→9	+13.9	271.4	+0.3	-13.9	+0.7
r50→26	+46.4	90.1	-0.1	+46.4	+0.2
r81→25	-22.8	315.4	+16.3	-16.0	-0.6
r82→8	+23.3	224.9	-16.5	-16.5	-0.3

max M +0.70 kN·m rod 49 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

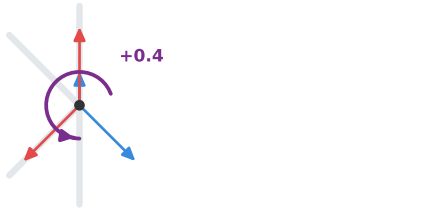
Joint 43



member	N	θ°	Fx	Fy	M
r51→10	-15.1	89.3	+0.2	+15.1	+0.5
r52→27	+74.5	89.9	+0.1	+74.5	-0.0
r83→26	-63.1	315.0	+44.7	-44.6	-0.4
r84→9	+63.6	225.1	-44.9	-45.0	-0.1

max M +0.48 kN·m rod 51 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

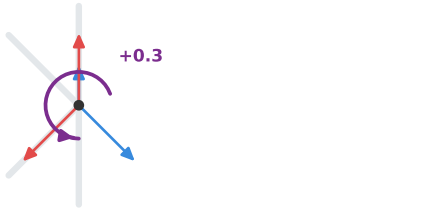
Joint 44



member	N	θ°	Fx	Fy	M
r53→11	-45.2	89.8	+0.1	+45.2	+0.4
r54→28	+104.8	89.9	+0.1	+104.8	-0.1
r85→27	-105.9	315.0	+74.9	-74.8	-0.3
r86→10	+106.3	225.0	-75.2	-75.2	+0.0

max M +0.40 kN·m rod 53 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

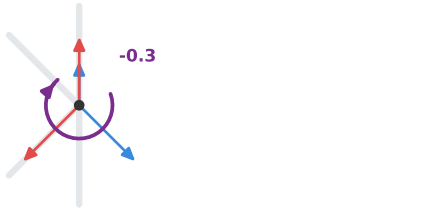
Joint 45



member	N	θ°	Fx	Fy	M
r55→12	-75.0	90.0	+0.0	+75.0	+0.3
r56→29	+134.7	89.9	+0.2	+134.7	-0.2
r87→28	-148.1	315.0	+104.7	-104.6	-0.2
r88→11	+148.5	225.0	-105.0	-105.0	+0.1

max M +0.27 kN·m rod 55 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

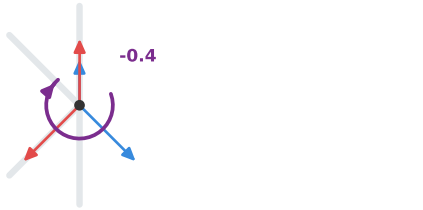
Joint 46



member	N	θ°	Fx	Fy	M
r57→13	-105.0	90.0	-0.1	+105.0	+0.1
r58→30	+164.4	89.9	+0.3	+164.4	-0.3
r89→29	-190.4	315.0	+134.7	-134.5	-0.1
r90→12	+190.7	225.0	-134.9	-134.9	+0.2

max M -0.28 kN·m rod 58 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 47

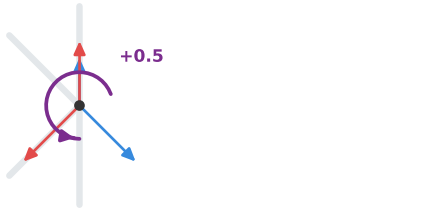


member	N	θ°	Fx	Fy	M
r59→14	-134.2	90.1	-0.2	+134.2	-0.0
r60→31	+195.2	89.9	+0.4	+195.2	-0.4
r91→30	-232.7	315.0	+164.6	-164.5	+0.1
r92→13	+233.1	225.0	-164.8	-164.8	+0.4

max M -0.41 kN·m rod 60 clockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), with its shear and end moment M at a rigid joint, the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 48



member	N	θ°	Fx	Fy	M
r61→15	-165.3	90.1	-0.3	+165.3	-0.3
r62→32	+223.2	89.9	+0.4	+223.2	-0.4
r93→31	-274.7	315.0	+194.3	-194.1	+0.1
r94→14	+274.9	225.0	-194.3	-194.4	+0.5

max M +0.53 kN·m rod 94 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Joint 49



member	N	θ°	Fx	Fy	M
r63→16	-448.1	90.2	-1.4	+448.1	-0.6
r64→33	-0.8	322.7	+1.1	-0.8	-0.7
r95→32	-316.5	315.0	+223.8	-223.8	+0.6
r96→15	+316.0	225.0	-223.5	-223.5	+0.7

max M +0.67 kN·m rod 96 anticlockwise
Σ kN Fx 0.00 Fy 0.00 M 0.00 ✓

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Bottom chord	HEA 220	4.00	+1.38	H1-1b, axial + bending	0.05 OK
1	1–2	Bottom chord	HEA 220	4.00	+225.18	normal stress (H.3.4)	0.17 OK
2	2–3	Bottom chord	HEA 220	4.00	+419.68	H1-1a, axial + bending	0.34 OK
3	3–4	Bottom chord	HEA 220	4.00	+584.55	H1-1a, axial + bending	0.47 OK
4	4–5	Bottom chord	HEA 220	4.00	+719.38	H1-1a, axial + bending	0.57 OK
5	5–6	Bottom chord	HEA 220	4.00	+824.23	H1-1a, axial + bending	0.65 OK
6	6–7	Bottom chord	HEA 220	4.00	+899.18	H1-1a, axial + bending	0.71 OK
7	7–8	Bottom chord	HEA 220	4.00	+943.77	H1-1a, axial + bending	0.75 OK
8	8–9	Bottom chord	HEA 220	4.00	+943.77	H1-1a, axial + bending	0.75 OK
9	9–10	Bottom chord	HEA 220	4.00	+899.18	H1-1a, axial + bending	0.71 OK
10	10–11	Bottom chord	HEA 220	4.00	+824.23	H1-1a, axial + bending	0.65 OK
11	11–12	Bottom chord	HEA 220	4.00	+719.38	H1-1a, axial + bending	0.57 OK
12	12–13	Bottom chord	HEA 220	4.00	+584.55	H1-1a, axial + bending	0.47 OK
13	13–14	Bottom chord	HEA 220	4.00	+419.68	H1-1a, axial + bending	0.34 OK
14	14–15	Bottom chord	HEA 220	4.00	+225.18	normal stress (H.3.4)	0.17 OK
15	15–16	Bottom chord	HEA 220	4.00	+1.38	H1-1b, axial + bending	0.05 OK
16	17–18	Top chord	HEB 220	4.00	-1.08	H1-1b, axial + bending	0.02 OK
17	18–19	Top chord	HEB 220	4.00	-225.22	inelastic buckling (E3-2)	0.15 OK
18	19–20	Top chord	HEB 220	4.00	-419.88	H1-1a, axial + bending	0.31 OK
19	20–21	Top chord	HEB 220	4.00	-584.75	H1-1a, axial + bending	0.42 OK
20	21–22	Top chord	HEB 220	4.00	-719.61	H1-1a, axial + bending	0.51 OK
21	22–23	Top chord	HEB 220	4.00	-824.47	H1-1a, axial + bending	0.58 OK
22	23–24	Top chord	HEB 220	4.00	-899.44	H1-1a, axial + bending	0.64 OK
23	24–25	Top chord	HEB 220	4.00	-944.03	H1-1a, axial + bending	0.67 OK
24	25–26	Top chord	HEB 220	4.00	-944.03	H1-1a, axial + bending	0.67 OK
25	26–27	Top chord	HEB 220	4.00	-899.44	H1-1a, axial + bending	0.64 OK
26	27–28	Top chord	HEB 220	4.00	-824.47	H1-1a, axial + bending	0.58 OK
27	28–29	Top chord	HEB 220	4.00	-719.61	H1-1a, axial + bending	0.51 OK
28	29–30	Top chord	HEB 220	4.00	-584.75	H1-1a, axial + bending	0.42 OK
29	30–31	Top chord	HEB 220	4.00	-419.88	H1-1a, axial + bending	0.31 OK
30	31–32	Top chord	HEB 220	4.00	-225.22	inelastic buckling (E3-2)	0.15 OK
31	32–33	Top chord	HEB 220	4.00	-1.08	H1-1b, axial + bending	0.02 OK
32	0–34	End post	SHS 200x200x8	4.00	-448.07	H1-1a, axial + bending	0.45 OK
33	34–17	End post	SHS 200x200x8	4.00	-0.82	H1-1b, axial + bending	0.05 OK
34	1–35	Vertical	SHS 120x120x5	4.00	-165.26	H1-1a, axial + bending	0.54 OK
35	35–18	Vertical	SHS 120x120x5	4.00	+223.25	H1-1a, axial + bending	0.51 OK
36	2–36	Vertical	SHS 120x120x5	4.00	-134.18	H1-1a, axial + bending	0.43 OK
37	36–19	Vertical	SHS 120x120x5	4.00	+195.15	H1-1a, axial + bending	0.45 OK
38	3–37	Vertical	SHS 120x120x5	4.00	-105.02	H1-1a, axial + bending	0.33 OK
39	37–20	Vertical	SHS 120x120x5	4.00	+164.41	H1-1a, axial + bending	0.38 OK
40	4–38	Vertical	SHS 120x120x5	4.00	-74.98	H1-1a, axial + bending	0.23 OK
41	38–21	Vertical	SHS 120x120x5	4.00	+134.67	H1-1a, axial + bending	0.31 OK
42	5–39	Vertical	SHS 120x120x5	4.00	-45.17	inelastic buckling (E3-2)	0.13 OK
43	39–22	Vertical	SHS 120x120x5	4.00	+104.81	H1-1a, axial + bending	0.23 OK
44	6–40	Vertical	SHS 120x120x5	4.00	-15.06	H1-1b, axial + bending	0.05 OK
45	40–23	Vertical	SHS 120x120x5	4.00	+74.52	normal stress (H.3.4)	0.15 OK
46	7–41	Vertical	SHS 120x120x5	4.00	+13.88	H1-1b, axial + bending	0.05 OK
47	41–24	Vertical	SHS 120x120x5	4.00	+46.36	normal stress (H.3.4)	0.10 OK
48	8–25	Vertical	SHS 120x120x5	8.00	+29.27	normal stress (H.3.4)	0.06 OK
49	9–42	Vertical	SHS 120x120x5	4.00	+13.88	H1-1b, axial + bending	0.05 OK
50	42–26	Vertical	SHS 120x120x5	4.00	+46.36	normal stress (H.3.4)	0.10 OK
51	10–43	Vertical	SHS 120x120x5	4.00	-15.06	H1-1b, axial + bending	0.05 OK

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
52	43–27	Vertical	SHS 120x120x5	4.00	+74.52	normal stress (H.3.4)	0.15 OK
53	11–44	Vertical	SHS 120x120x5	4.00	-45.17	inelastic buckling (E3-2)	0.13 OK
54	44–28	Vertical	SHS 120x120x5	4.00	+104.81	H1-1a, axial + bending	0.23 OK
55	12–45	Vertical	SHS 120x120x5	4.00	-74.98	H1-1a, axial + bending	0.23 OK
56	45–29	Vertical	SHS 120x120x5	4.00	+134.67	H1-1a, axial + bending	0.31 OK
57	13–46	Vertical	SHS 120x120x5	4.00	-105.02	H1-1a, axial + bending	0.33 OK
58	46–30	Vertical	SHS 120x120x5	4.00	+164.41	H1-1a, axial + bending	0.38 OK
59	14–47	Vertical	SHS 120x120x5	4.00	-134.18	H1-1a, axial + bending	0.43 OK
60	47–31	Vertical	SHS 120x120x5	4.00	+195.15	H1-1a, axial + bending	0.45 OK
61	15–48	Vertical	SHS 120x120x5	4.00	-165.26	H1-1a, axial + bending	0.54 OK
62	48–32	Vertical	SHS 120x120x5	4.00	+223.25	H1-1a, axial + bending	0.51 OK
63	16–49	End post	SHS 200x200x8	4.00	-448.07	H1-1a, axial + bending	0.45 OK
64	49–33	End post	SHS 200x200x8	4.00	-0.82	H1-1b, axial + bending	0.05 OK
65	34–18	Diagonal	SHS 150x150x6	5.66	-316.46	H1-1a, axial + bending	0.70 OK
66	34–1	Diagonal	SHS 150x150x6	5.66	+316.03	H1-1a, axial + bending	0.45 OK
67	35–19	Diagonal	SHS 150x150x6	5.66	-274.65	H1-1a, axial + bending	0.62 OK
68	35–2	Diagonal	SHS 150x150x6	5.66	+274.86	H1-1a, axial + bending	0.39 OK
69	36–20	Diagonal	SHS 150x150x6	5.66	-232.71	H1-1a, axial + bending	0.52 OK
70	36–3	Diagonal	SHS 150x150x6	5.66	+233.09	H1-1a, axial + bending	0.33 OK
71	37–21	Diagonal	SHS 150x150x6	5.66	-190.35	H1-1a, axial + bending	0.42 OK
72	37–4	Diagonal	SHS 150x150x6	5.66	+190.74	H1-1a, axial + bending	0.27 OK
73	38–22	Diagonal	SHS 150x150x6	5.66	-148.05	H1-1a, axial + bending	0.33 OK
74	38–5	Diagonal	SHS 150x150x6	5.66	+148.48	H1-1a, axial + bending	0.21 OK
75	39–23	Diagonal	SHS 150x150x6	5.66	-105.86	H1-1a, axial + bending	0.24 OK
76	39–6	Diagonal	SHS 150x150x6	5.66	+106.31	normal stress (H.3.4)	0.15 OK
77	40–24	Diagonal	SHS 150x150x6	5.66	-63.10	inelastic buckling (E3-2)	0.14 OK
78	40–7	Diagonal	SHS 150x150x6	5.66	+63.57	normal stress (H.3.4)	0.09 OK
79	41–25	Diagonal	SHS 150x150x6	5.66	-22.84	inelastic buckling (E3-2)	0.05 OK
80	41–8	Diagonal	SHS 150x150x6	5.66	+23.32	normal stress (H.3.4)	0.03 OK
81	42–25	Diagonal	SHS 150x150x6	5.66	-22.84	inelastic buckling (E3-2)	0.05 OK
82	42–8	Diagonal	SHS 150x150x6	5.66	+23.32	normal stress (H.3.4)	0.03 OK
83	43–26	Diagonal	SHS 150x150x6	5.66	-63.10	inelastic buckling (E3-2)	0.14 OK
84	43–9	Diagonal	SHS 150x150x6	5.66	+63.57	normal stress (H.3.4)	0.09 OK
85	44–27	Diagonal	SHS 150x150x6	5.66	-105.86	H1-1a, axial + bending	0.24 OK
86	44–10	Diagonal	SHS 150x150x6	5.66	+106.31	normal stress (H.3.4)	0.15 OK
87	45–28	Diagonal	SHS 150x150x6	5.66	-148.05	H1-1a, axial + bending	0.33 OK
88	45–11	Diagonal	SHS 150x150x6	5.66	+148.48	H1-1a, axial + bending	0.21 OK
89	46–29	Diagonal	SHS 150x150x6	5.66	-190.35	H1-1a, axial + bending	0.42 OK
90	46–12	Diagonal	SHS 150x150x6	5.66	+190.74	H1-1a, axial + bending	0.27 OK
91	47–30	Diagonal	SHS 150x150x6	5.66	-232.71	H1-1a, axial + bending	0.52 OK
92	47–13	Diagonal	SHS 150x150x6	5.66	+233.09	H1-1a, axial + bending	0.33 OK
93	48–31	Diagonal	SHS 150x150x6	5.66	-274.65	H1-1a, axial + bending	0.62 OK
94	48–14	Diagonal	SHS 150x150x6	5.66	+274.86	H1-1a, axial + bending	0.39 OK
95	49–32	Diagonal	SHS 150x150x6	5.66	-316.46	H1-1a, axial + bending	0.70 OK
96	49–15	Diagonal	SHS 150x150x6	5.66	+316.03	H1-1a, axial + bending	0.45 OK

Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 0

joint 0 → 1

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +1.38 kN
at 0:  Fx = +1.38   Fy = +0.00
at 1:  Fx = -1.38   Fy = +0.00
rigid: V = +1.93  Ma = -4.94  Mb = -2.77
```

joint 0 joint 1



Rod 1

joint 1 → 2

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +225.18 kN
at 1:  Fx = +225.18   Fy = +0.00
at 2:  Fx = -225.18   Fy = +0.00
rigid: V = +0.14  Ma = +1.06  Mb = -1.60
```

joint 1 joint 2



Rod 2

joint 2 → 3

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +419.68 kN
at 2:  Fx = +419.68   Fy = +0.00
at 3:  Fx = -419.68   Fy = +0.00
rigid: V = +0.34  Ma = +0.61  Mb = -1.97
```

joint 2 joint 3



Rod 3

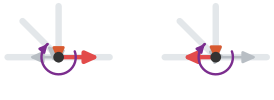
joint 3 → 4

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +584.55 kN
at 3:  Fx = +584.55   Fy = +0.00
at 4:  Fx = -584.55   Fy = +0.00
rigid: V = +0.15  Ma = +1.33  Mb = -1.94
```

joint 3 joint 4



Rod 4

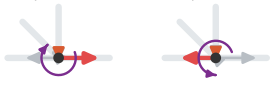
joint 4 → 5

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +719.38 kN
at 4:  Fx = +719.38   Fy = +0.00
at 5:  Fx = -719.38   Fy = +0.00
rigid: V = +0.07  Ma = +1.71  Mb = -1.99
```

joint 4 joint 5



Rod 5

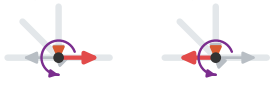
joint 5 → 6

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +824.23 kN
at 5:  Fx = +824.23   Fy = +0.00
at 6:  Fx = -824.23   Fy = +0.00
rigid: V = -0.09  Ma = +2.14  Mb = -1.79
```

joint 5 joint 6



Rod 6

joint 6 → 7

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +899.18 kN
at 6:  Fx = +899.18   Fy = +0.00
at 7:  Fx = -899.18   Fy = +0.00
rigid: V = +0.05  Ma = +2.23  Mb = -2.41
```

joint 6 joint 7



Rod 7

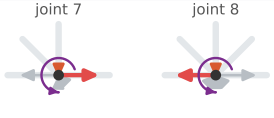
joint 7 → 8

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +943.77 kN
at 7:  Fx = +943.77   Fy = +0.00
at 8:  Fx = -943.77   Fy = +0.00
rigid: V = -1.08  Ma = +3.41  Mb = +0.93
```

joint 7 joint 8



Rod 8

joint 8 → 9

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +943.77 kN
at 8:  Fx = +943.77   Fy = +0.00
at 9:  Fx = -943.77   Fy = +0.00
rigid: V = +1.08  Ma = -0.93  Mb = -3.41
```

joint 8 joint 9



Rod 9

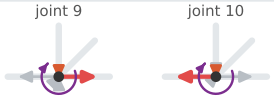
joint 9 → 10

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +899.18 kN
at 9:  Fx = +899.18   Fy = +0.00
at 10:  Fx = -899.18   Fy = +0.00
rigid: V = -0.05  Ma = +2.41  Mb = -2.23
```

joint 9 joint 10



Rod 10

joint 10 → 11

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +824.23 kN
at 10:  Fx = +824.23   Fy = +0.00
at 11:  Fx = -824.23   Fy = +0.00
rigid: V = +0.09  Ma = +1.79  Mb = -2.14
```

joint 10 joint 11



Rod 11

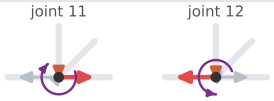
joint 11 → 12

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +719.38 kN
at 11:  Fx = +719.38   Fy = +0.00
at 12:  Fx = -719.38   Fy = +0.00
rigid: V = -0.07  Ma = +1.99  Mb = -1.71
```

joint 11 joint 12



Rod 12

joint 12 → 13

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +584.55 kN
at 12:  Fx = +584.55   Fy = +0.00
at 13:  Fx = -584.55   Fy = +0.00
rigid: V = -0.15  Ma = +1.94  Mb = -1.33
```

joint 12 joint 13



Rod 13

joint 13 → 14

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +419.68 kN
at 13:  Fx = +419.68   Fy = +0.00
at 14:  Fx = -419.68   Fy = +0.00
rigid: V = -0.34  Ma = +1.97  Mb = -0.61
```

joint 13 joint 14



Rod 14

joint 14 → 15

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +225.18 kN
at 14:  Fx = +225.18   Fy = +0.00
at 15:  Fx = -225.18   Fy = +0.00
rigid: V = -0.14  Ma = +1.60  Mb = -1.06
```

joint 14 joint 15



Rod 15

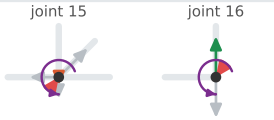
joint 15 → 16

tension



```
L = 4.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +1.38 kN
at 15:  Fx = +1.38   Fy = +0.00
at 16:  Fx = -1.38   Fy = +0.00
rigid: V = -1.93  Ma = +2.77  Mb = +4.94
```

joint 15 joint 16



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 16

joint 17 → 18

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -1.08 kN
at 17: Fx = -1.08 Fy = +0.00
at 18: Fx = +1.08 Fy = +0.00
rigid: V = +0.82 Ma = -3.67 Mb = +0.37

joint 17



joint 18



Rod 17

joint 18 → 19

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -225.22 kN
at 18: Fx = -225.22 Fy = +0.00
at 19: Fx = +225.22 Fy = +0.00
rigid: V = +1.35 Ma = -1.97 Mb = -3.45

joint 18



joint 19



Rod 18

joint 19 → 20

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -419.88 kN
at 19: Fx = -419.88 Fy = +0.00
at 20: Fx = +419.88 Fy = +0.00
rigid: V = +0.33 Ma = +1.61 Mb = -2.94

joint 19



joint 20



Rod 19

joint 20 → 21

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -584.75 kN
at 20: Fx = -584.75 Fy = +0.00
at 21: Fx = +584.75 Fy = +0.00
rigid: V = +0.41 Ma = +1.62 Mb = -3.27

joint 20



joint 21



Rod 20

joint 21 → 22

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -719.61 kN
at 21: Fx = -719.61 Fy = +0.00
at 22: Fx = +719.61 Fy = +0.00
rigid: V = +0.28 Ma = +2.28 Mb = -3.40

joint 21



joint 22



Rod 21

joint 22 → 23

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -824.47 kN
at 22: Fx = -824.47 Fy = +0.00
at 23: Fx = +824.47 Fy = +0.00
rigid: V = +0.10 Ma = +2.77 Mb = -3.19

joint 22



joint 23



Rod 22

joint 23 → 24

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -899.44 kN
at 23: Fx = -899.44 Fy = +0.00
at 24: Fx = +899.44 Fy = +0.00
rigid: V = +0.37 Ma = +2.82 Mb = -4.32

joint 23



joint 24



Rod 23

joint 24 → 25

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -944.03 kN
at 24: Fx = -944.03 Fy = +0.00
at 25: Fx = +944.03 Fy = +0.00
rigid: V = -1.40 Ma = +4.54 Mb = +1.04

joint 24



joint 25



Rod 24

joint 25 → 26

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -944.03 kN
at 25: Fx = -944.03 Fy = +0.00
at 26: Fx = +944.03 Fy = +0.00
rigid: V = +1.40 Ma = -1.04 Mb = -4.54

joint 25



joint 26



Rod 25

joint 26 → 27

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -899.44 kN
at 26: Fx = -899.44 Fy = +0.00
at 27: Fx = +899.44 Fy = +0.00
rigid: V = -0.37 Ma = +4.32 Mb = -2.82

joint 26



joint 27



Rod 26

joint 27 → 28

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -824.47 kN
at 27: Fx = -824.47 Fy = +0.00
at 28: Fx = +824.47 Fy = +0.00
rigid: V = -0.10 Ma = +3.19 Mb = -2.77

joint 27



joint 28



Rod 27

joint 28 → 29

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -719.61 kN
at 28: Fx = -719.61 Fy = +0.00
at 29: Fx = +719.61 Fy = +0.00
rigid: V = -0.28 Ma = +3.40 Mb = -2.28

joint 28



joint 29



Rod 28

joint 29 → 30

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -584.75 kN
at 29: Fx = -584.75 Fy = +0.00
at 30: Fx = +584.75 Fy = +0.00
rigid: V = -0.41 Ma = +3.27 Mb = -1.62

joint 29



joint 30



Rod 29

joint 30 → 31

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -419.88 kN
at 30: Fx = -419.88 Fy = +0.00
at 31: Fx = +419.88 Fy = +0.00
rigid: V = -0.33 Ma = +2.94 Mb = -1.61

joint 30



joint 31



Rod 30

joint 31 → 32

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -225.22 kN
at 31: Fx = -225.22 Fy = +0.00
at 32: Fx = +225.22 Fy = +0.00
rigid: V = -1.35 Ma = +3.45 Mb = +1.97

joint 31



joint 32



Rod 31

joint 32 → 33

compression



L = 4.000 m θ = 0.0°
cos θ = +1.000 sin θ = +0.000
N = -1.08 kN
at 32: Fx = -1.08 Fy = +0.00
at 33: Fx = +1.08 Fy = +0.00
rigid: V = -0.82 Ma = -0.37 Mb = +3.67

joint 32



joint 33



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 32

joint 0 → 34

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -448.07 kN
at 0:   Fx = +0.00   Fy = -448.07
at 34:  Fx = +0.00   Fy = +448.07
rigid: V = -1.38  Ma = +4.94  Mb = +0.58
```

joint 0



joint 34



Rod 33

joint 34 → 17

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -0.82 kN
at 34:  Fx = +0.00   Fy = -0.82
at 17:  Fx = +0.00   Fy = +0.82
rigid: V = -1.08  Ma = +0.67  Mb = +3.67
```

joint 34



joint 17



Rod 34

joint 1 → 35

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -165.26 kN
at 1:   Fx = +0.00   Fy = -165.26
at 35:  Fx = +0.00   Fy = +165.26
rigid: V = -0.33  Ma = +1.06  Mb = +0.27
```

joint 1



joint 35



Rod 35

joint 35 → 18

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +223.25 kN
at 35:  Fx = +0.00   Fy = +223.25
at 18:  Fx = +0.00   Fy = -223.25
rigid: V = -0.37  Ma = +0.42  Mb = +1.07
```

joint 35



joint 18



Rod 36

joint 2 → 36

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -134.18 kN
at 2:   Fx = +0.00   Fy = -134.18
at 36:  Fx = +0.00   Fy = +134.18
rigid: V = -0.17  Ma = +0.65  Mb = +0.03
```

joint 2



joint 36



Rod 37

joint 36 → 19

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +195.15 kN
at 36:  Fx = +0.00   Fy = +195.15
at 19:  Fx = +0.00   Fy = -195.15
rigid: V = -0.37  Ma = +0.41  Mb = +1.06
```

joint 36



joint 19



Rod 38

joint 3 → 37

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -105.02 kN
at 3:   Fx = +0.00   Fy = -105.02
at 37:  Fx = +0.00   Fy = +105.02
rigid: V = -0.07  Ma = +0.40  Mb = -0.12
```

joint 3



joint 37



Rod 39

joint 37 → 20

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +164.41 kN
at 37:  Fx = +0.00   Fy = +164.41
at 20:  Fx = +0.00   Fy = -164.41
rigid: V = -0.27  Ma = +0.28  Mb = +0.79
```

joint 37



joint 20



Rod 40

joint 4 → 38

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -74.98 kN
at 4:   Fx = +0.00   Fy = -74.98
at 38:  Fx = +0.00   Fy = +74.98
rigid: V = +0.03  Ma = +0.14  Mb = -0.27
```

joint 4



joint 38



Rod 41

joint 38 → 21

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +134.67 kN
at 38:  Fx = +0.00   Fy = +134.67
at 21:  Fx = +0.00   Fy = -134.67
rigid: V = -0.20  Ma = +0.19  Mb = +0.59
```

joint 38



joint 21



Rod 42

joint 5 → 39

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -45.17 kN
at 5:   Fx = +0.00   Fy = -45.17
at 39:  Fx = +0.00   Fy = +45.17
rigid: V = +0.13  Ma = -0.10  Mb = -0.40
```

joint 5



joint 39



Rod 43

joint 39 → 22

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +104.81 kN
at 39:  Fx = +0.00   Fy = +104.81
at 22:  Fx = +0.00   Fy = -104.81
rigid: V = -0.12  Ma = +0.09  Mb = +0.38
```

joint 39



joint 22



Rod 44

joint 6 → 40

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -15.06 kN
at 6:   Fx = +0.00   Fy = -15.06
at 40:  Fx = +0.00   Fy = +15.06
rigid: V = +0.19  Ma = -0.29  Mb = -0.48
```

joint 6



joint 40



Rod 45

joint 40 → 23

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +74.52 kN
at 40:  Fx = +0.00   Fy = +74.52
at 23:  Fx = +0.00   Fy = -74.52
rigid: V = -0.07  Ma = +0.04  Mb = +0.22
```

joint 40



joint 23



Rod 46

joint 7 → 41

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +13.88 kN
at 7:   Fx = +0.00   Fy = +13.88
at 41:  Fx = +0.00   Fy = -13.88
rigid: V = +0.33  Ma = -0.62  Mb = -0.70
```

joint 7



joint 41



Rod 47

joint 41 → 24

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +46.36 kN
at 41:  Fx = +0.00   Fy = +46.36
at 24:  Fx = +0.00   Fy = -46.36
rigid: V = +0.07  Ma = -0.18  Mb = -0.11
```

joint 41



joint 24



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 48

joint 8 → 25

tension



```
L = 8.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +29.27 kN
at 8:   Fx = +0.00   Fy = +29.27
at 25:  Fx = +0.00   Fy = -29.27
rigid: V = +0.00  Ma = +0.00  Mb = +0.00
```

joint 8 joint 25



Rod 49

joint 9 → 42

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +13.88 kN
at 9:   Fx = +0.00   Fy = +13.88
at 42:  Fx = +0.00   Fy = -13.88
rigid: V = -0.33  Ma = +0.62  Mb = +0.70
```

joint 9 joint 42



Rod 50

joint 42 → 26

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +46.36 kN
at 42:  Fx = +0.00   Fy = +46.36
at 26:  Fx = +0.00   Fy = -46.36
rigid: V = -0.07  Ma = +0.18  Mb = +0.11
```

joint 42 joint 26



Rod 51

joint 10 → 43

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -15.06 kN
at 10:  Fx = +0.00   Fy = -15.06
at 43:  Fx = +0.00   Fy = +15.06
rigid: V = -0.19  Ma = +0.29  Mb = +0.48
```

joint 10 joint 43



Rod 52

joint 43 → 27

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +74.52 kN
at 43:  Fx = +0.00   Fy = +74.52
at 27:  Fx = +0.00   Fy = -74.52
rigid: V = +0.07  Ma = -0.04  Mb = -0.22
```

joint 43 joint 27



Rod 53

joint 11 → 44

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -45.17 kN
at 11:  Fx = +0.00   Fy = -45.17
at 44:  Fx = +0.00   Fy = +45.17
rigid: V = -0.13  Ma = +0.10  Mb = +0.40
```

joint 11 joint 44



Rod 54

joint 44 → 28

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +104.81 kN
at 44:  Fx = +0.00   Fy = +104.81
at 28:  Fx = +0.00   Fy = -104.81
rigid: V = +0.12  Ma = -0.09  Mb = -0.38
```

joint 44 joint 28



Rod 55

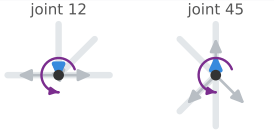
joint 12 → 45

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -74.98 kN
at 12:  Fx = +0.00   Fy = -74.98
at 45:  Fx = +0.00   Fy = +74.98
rigid: V = -0.03  Ma = -0.14  Mb = +0.27
```

joint 12 joint 45



Rod 56

joint 45 → 29

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +134.67 kN
at 45:  Fx = +0.00   Fy = +134.67
at 29:  Fx = +0.00   Fy = -134.67
rigid: V = +0.20  Ma = -0.19  Mb = -0.59
```

joint 45 joint 29



Rod 57

joint 13 → 46

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -105.02 kN
at 13:  Fx = +0.00   Fy = -105.02
at 46:  Fx = +0.00   Fy = +105.02
rigid: V = +0.07  Ma = -0.40  Mb = +0.12
```

joint 13 joint 46



Rod 58

joint 46 → 30

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +164.41 kN
at 46:  Fx = +0.00   Fy = +164.41
at 30:  Fx = +0.00   Fy = -164.41
rigid: V = +0.27  Ma = -0.28  Mb = -0.79
```

joint 46 joint 30



Rod 59

joint 14 → 47

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -134.18 kN
at 14:  Fx = +0.00   Fy = -134.18
at 47:  Fx = +0.00   Fy = +134.18
rigid: V = +0.17  Ma = -0.65  Mb = -0.03
```

joint 14 joint 47



Rod 60

joint 47 → 31

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +195.15 kN
at 47:  Fx = +0.00   Fy = +195.15
at 31:  Fx = +0.00   Fy = -195.15
rigid: V = +0.37  Ma = -0.41  Mb = -1.06
```

joint 47 joint 31



Rod 61

joint 15 → 48

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -165.26 kN
at 15:  Fx = +0.00   Fy = -165.26
at 48:  Fx = +0.00   Fy = +165.26
rigid: V = +0.33  Ma = -1.06  Mb = -0.27
```

joint 15 joint 48



Rod 62

joint 48 → 32

tension



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +223.25 kN
at 48:  Fx = +0.00   Fy = +223.25
at 32:  Fx = +0.00   Fy = -223.25
rigid: V = +0.37  Ma = -0.42  Mb = -1.07
```

joint 48 joint 32



Rod 63

joint 16 → 49

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -448.07 kN
at 16:  Fx = +0.00   Fy = -448.07
at 49:  Fx = +0.00   Fy = +448.07
rigid: V = +1.38  Ma = -4.94  Mb = -0.58
```

joint 16 joint 49



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 64

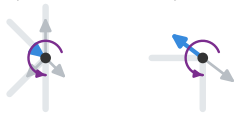
joint 49 → 33

compression



```
L = 4.000 m      θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = -0.82 kN
at 49:  Fx = +0.00   Fy = -0.82
at 33:  Fx = +0.00   Fy = +0.82
rigid: V = +1.08  Ma = -0.67  Mb = -3.67
```

joint 49 joint 33



Rod 65

joint 34 → 18

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -316.46 kN
at 34:  Fx = -223.77   Fy = -223.77
at 18:  Fx = +223.77   Fy = +223.77
rigid: V = +0.01  Ma = -0.58  Mb = +0.53
```

joint 34 joint 18



Rod 66

joint 34 → 1

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +316.03 kN
at 34:  Fx = +223.47   Fy = -223.47
at 1:  Fx = -223.47   Fy = +223.47
rigid: V = +0.00  Ma = -0.67  Mb = +0.66
```

joint 34 joint 1



Rod 67

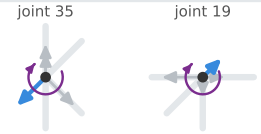
joint 35 → 19

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -274.65 kN
at 35:  Fx = -194.21   Fy = -194.21
at 19:  Fx = +194.21   Fy = +194.21
rigid: V = -0.11  Ma = -0.15  Mb = +0.78
```

joint 35 joint 19



Rod 68

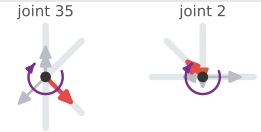
joint 35 → 2

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +274.86 kN
at 35:  Fx = +194.35   Fy = -194.35
at 2:  Fx = -194.35   Fy = +194.35
rigid: V = +0.03  Ma = -0.53  Mb = +0.35
```

joint 35 joint 2



Rod 69

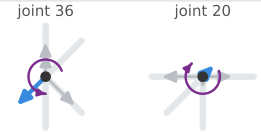
joint 36 → 20

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -232.71 kN
at 36:  Fx = -164.55   Fy = -164.55
at 20:  Fx = +164.55   Fy = +164.55
rigid: V = -0.08  Ma = -0.06  Mb = +0.53
```

joint 36 joint 20



Rod 70

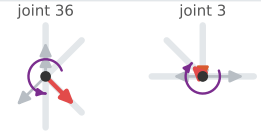
joint 36 → 3

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +233.09 kN
at 36:  Fx = +164.82   Fy = -164.82
at 3:  Fx = -164.82   Fy = +164.82
rigid: V = +0.02  Ma = -0.37  Mb = +0.24
```

joint 36 joint 3



Rod 71

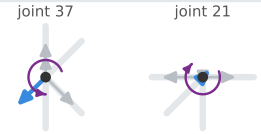
joint 37 → 21

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -190.35 kN
at 37:  Fx = -134.60   Fy = -134.60
at 21:  Fx = +134.60   Fy = +134.60
rigid: V = -0.09  Ma = +0.08  Mb = +0.41
```

joint 37 joint 21



Rod 72

joint 37 → 4

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +190.74 kN
at 37:  Fx = +134.88   Fy = -134.88
at 4:  Fx = -134.88   Fy = +134.88
rigid: V = +0.02  Ma = -0.24  Mb = +0.10
```

joint 37 joint 4



Rod 73

joint 38 → 22

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -148.05 kN
at 38:  Fx = -104.69   Fy = -104.69
at 22:  Fx = +104.69   Fy = +104.69
rigid: V = -0.08  Ma = +0.20  Mb = +0.25
```

joint 38 joint 22



Rod 74

joint 38 → 5

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +148.48 kN
at 38:  Fx = +104.99   Fy = -104.99
at 5:  Fx = -104.99   Fy = +104.99
rigid: V = +0.03  Ma = -0.12  Mb = -0.04
```

joint 38 joint 5



Rod 75

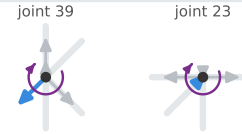
joint 39 → 23

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -105.86 kN
at 39:  Fx = -74.85   Fy = -74.85
at 23:  Fx = +74.85   Fy = +74.85
rigid: V = -0.08  Ma = +0.31  Mb = +0.15
```

joint 39 joint 23



Rod 76

joint 39 → 6

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +106.31 kN
at 39:  Fx = +75.17   Fy = -75.17
at 6:  Fx = -75.17   Fy = +75.17
rigid: V = +0.03  Ma = +0.00  Mb = -0.15
```

joint 39 joint 6



Rod 77

joint 40 → 24

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -63.10 kN
at 40:  Fx = -44.62   Fy = -44.62
at 24:  Fx = +44.62   Fy = +44.62
rigid: V = -0.05  Ma = +0.37  Mb = -0.11
```

joint 40 joint 24



Rod 78

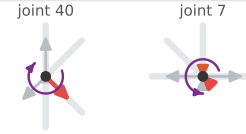
joint 40 → 7

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +63.57 kN
at 40:  Fx = +44.95   Fy = -44.95
at 7:  Fx = -44.95   Fy = +44.95
rigid: V = +0.06  Ma = +0.07  Mb = -0.38
```

joint 40 joint 7



Rod 79

joint 41 → 25

compression



```
L = 5.657 m      θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -22.84 kN
at 41:  Fx = -16.15   Fy = -16.15
at 25:  Fx = +16.15   Fy = +16.15
rigid: V = -0.17  Ma = +0.60  Mb = +0.34
```

joint 41 joint 25



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 80

joint 41 → 8

tension



```
L = 5.657 m      θ = 315.0°
cos θ = +0.707   sin θ = -0.707
N = +23.32 kN
at 41:  Fx = +16.49   Fy = -16.49
at 8:   Fx = -16.49   Fy = +16.49
rigid: V = -0.05  Ma = +0.28  Mb = +0.02
```

joint 41 joint 8



Rod 81

joint 42 → 25

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -22.84 kN
at 42:  Fx = +16.15   Fy = -16.15
at 25:  Fx = -16.15   Fy = +16.15
rigid: V = +0.17  Ma = -0.60  Mb = -0.34
```

joint 42 joint 25



Rod 82

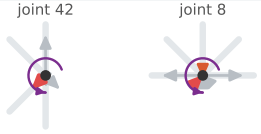
joint 42 → 8

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +23.32 kN
at 42:  Fx = -16.49   Fy = -16.49
at 8:   Fx = +16.49   Fy = +16.49
rigid: V = +0.05  Ma = -0.28  Mb = -0.02
```

joint 42 joint 8



Rod 83

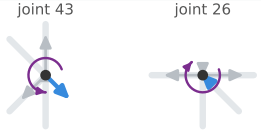
joint 43 → 26

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -63.10 kN
at 43:  Fx = +44.62   Fy = -44.62
at 26:  Fx = -44.62   Fy = +44.62
rigid: V = +0.05  Ma = -0.37  Mb = +0.11
```

joint 43 joint 26



Rod 84

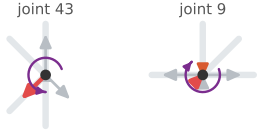
joint 43 → 9

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +63.57 kN
at 43:  Fx = -44.95   Fy = -44.95
at 9:   Fx = +44.95   Fy = +44.95
rigid: V = -0.06  Ma = -0.07  Mb = +0.38
```

joint 43 joint 9



Rod 85

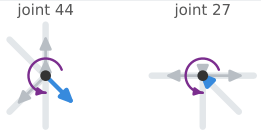
joint 44 → 27

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -105.86 kN
at 44:  Fx = +74.85   Fy = -74.85
at 27:  Fx = -74.85   Fy = +74.85
rigid: V = +0.08  Ma = -0.31  Mb = -0.15
```

joint 44 joint 27



Rod 86

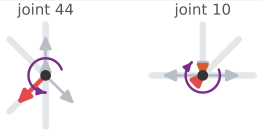
joint 44 → 10

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +106.31 kN
at 44:  Fx = -75.17   Fy = -75.17
at 10:  Fx = +75.17   Fy = +75.17
rigid: V = -0.03  Ma = +0.00  Mb = +0.15
```

joint 44 joint 10



Rod 87

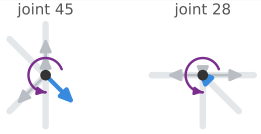
joint 45 → 28

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -148.05 kN
at 45:  Fx = +104.69   Fy = -104.69
at 28:  Fx = -104.69   Fy = +104.69
rigid: V = +0.08  Ma = -0.20  Mb = -0.25
```

joint 45 joint 28



Rod 88

joint 45 → 11

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +148.48 kN
at 45:  Fx = -104.99   Fy = -104.99
at 11:  Fx = +104.99   Fy = +104.99
rigid: V = -0.03  Ma = +0.12  Mb = +0.04
```

joint 45 joint 11



Rod 89

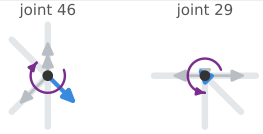
joint 46 → 29

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -190.35 kN
at 46:  Fx = +134.60   Fy = -134.60
at 29:  Fx = -134.60   Fy = +134.60
rigid: V = +0.09  Ma = -0.08  Mb = -0.41
```

joint 46 joint 29



Rod 90

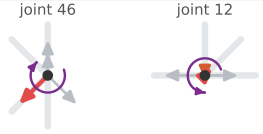
joint 46 → 12

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +190.74 kN
at 46:  Fx = -134.88   Fy = -134.88
at 12:  Fx = +134.88   Fy = +134.88
rigid: V = -0.02  Ma = +0.24  Mb = -0.10
```

joint 46 joint 12



Rod 91

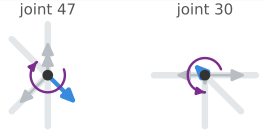
joint 47 → 30

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -232.71 kN
at 47:  Fx = +164.55   Fy = -164.55
at 30:  Fx = -164.55   Fy = +164.55
rigid: V = +0.08  Ma = +0.06  Mb = -0.53
```

joint 47 joint 30



Rod 92

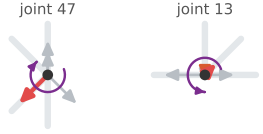
joint 47 → 13

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +233.09 kN
at 47:  Fx = -164.82   Fy = -164.82
at 13:  Fx = +164.82   Fy = +164.82
rigid: V = -0.02  Ma = +0.37  Mb = -0.24
```

joint 47 joint 13



Rod 93

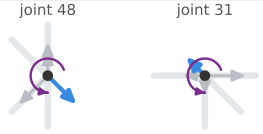
joint 48 → 31

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -274.65 kN
at 48:  Fx = +194.21   Fy = -194.21
at 31:  Fx = -194.21   Fy = +194.21
rigid: V = +0.11  Ma = +0.15  Mb = -0.78
```

joint 48 joint 31



Rod 94

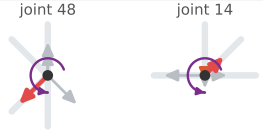
joint 48 → 14

tension



```
L = 5.657 m      θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +274.86 kN
at 48:  Fx = -194.35   Fy = -194.35
at 14:  Fx = +194.35   Fy = +194.35
rigid: V = -0.03  Ma = +0.53  Mb = -0.35
```

joint 48 joint 14



Rod 95

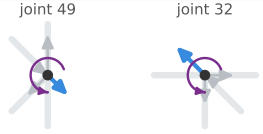
joint 49 → 32

compression



```
L = 5.657 m      θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -316.46 kN
at 49:  Fx = +223.77   Fy = -223.77
at 32:  Fx = -223.77   Fy = +223.77
rigid: V = -0.01  Ma = +0.58  Mb = -0.53
```

joint 49 joint 32



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 96

joint 49 → 15

tension



```
L = 5.657 m     θ = 225.0°
cos θ = -0.707   sin θ = -0.707
N = +316.03 kN
at 49:  Fx = -223.47   Fy = -223.47
at 15:  Fx = +223.47   Fy = +223.47
rigid: V = +0.00  Ma = +0.67  Mb = -0.66
```

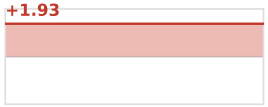
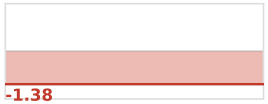
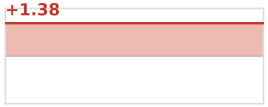
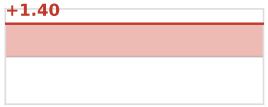
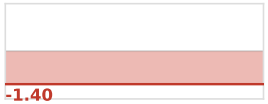
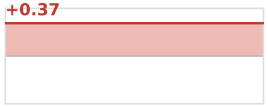
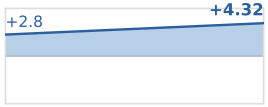
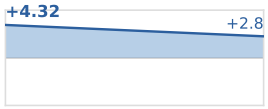
joint 49

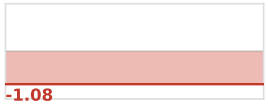
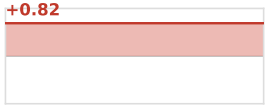
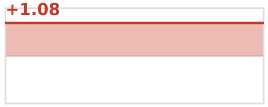
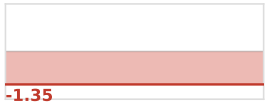
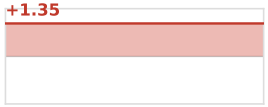
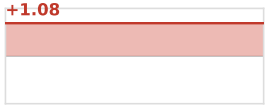


joint 15

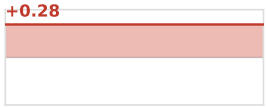
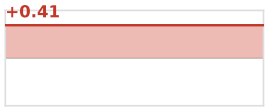
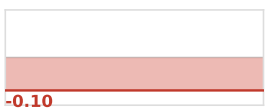
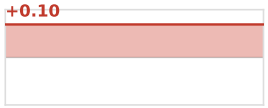
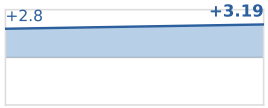
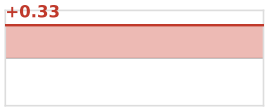
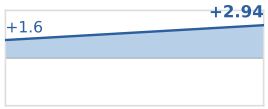


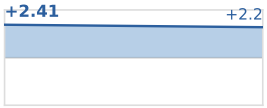
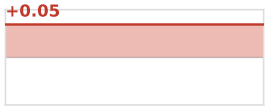
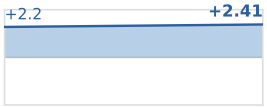
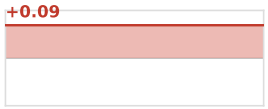
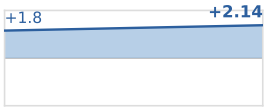
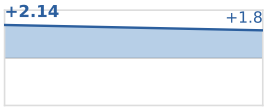
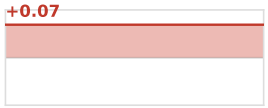
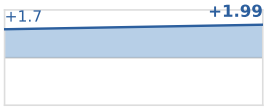
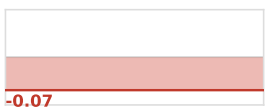
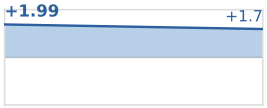
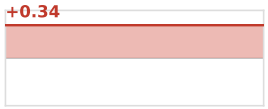
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 0 use 0.05 0–1 · 4.00 m HEA 220 N +1.38 kN			
Rod 32 use 0.45 0–34 · 4.00 m SHS 200x200x8 N -448.07 kN			
Rod 63 use 0.45 16–49 · 4.00 m SHS 200x200x8 N -448.07 kN			
Rod 15 use 0.05 15–16 · 4.00 m HEA 220 N +1.38 kN			
Rod 24 use 0.67 25–26 · 4.00 m HEB 220 N -944.03 kN			
Rod 23 use 0.67 24–25 · 4.00 m HEB 220 N -944.03 kN			
Rod 22 use 0.64 23–24 · 4.00 m HEB 220 N -899.44 kN			
Rod 25 use 0.64 26–27 · 4.00 m HEB 220 N -899.44 kN			

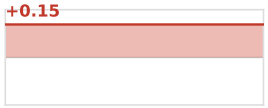
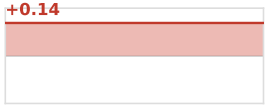
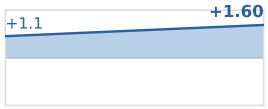
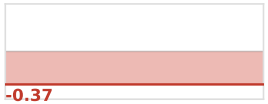
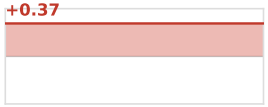
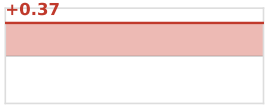
	Where	Shear V (kN)	Moment M (kN·m)
Rod 31 use 0.02 32–33 · 4.00 m HEB 220 N -1.08 kN			
Rod 33 use 0.05 34–17 · 4.00 m SHS 200x200x8 N -0.82 kN			
Rod 16 use 0.02 17–18 · 4.00 m HEB 220 N -1.08 kN			
Rod 64 use 0.05 49–33 · 4.00 m SHS 200x200x8 N -0.82 kN			
Rod 30 use 0.15 31–32 · 4.00 m HEB 220 N -225.22 kN			
Rod 17 use 0.15 18–19 · 4.00 m HEB 220 N -225.22 kN			
Rod 8 use 0.75 8–9 · 4.00 m HEA 220 N +943.77 kN			
Rod 7 use 0.75 7–8 · 4.00 m HEA 220 N +943.77 kN			

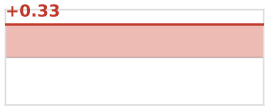
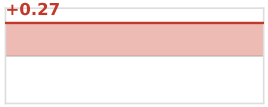
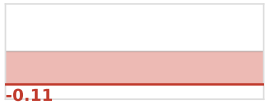
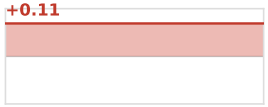
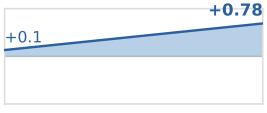
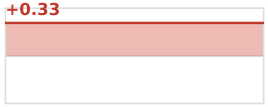
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 27 use 0.51 28–29 · 4.00 m HEB 220 N -719.61 kN			
Rod 20 use 0.51 21–22 · 4.00 m HEB 220 N -719.61 kN			
Rod 19 use 0.42 20–21 · 4.00 m HEB 220 N -584.75 kN			
Rod 28 use 0.42 29–30 · 4.00 m HEB 220 N -584.75 kN			
Rod 26 use 0.58 27–28 · 4.00 m HEB 220 N -824.47 kN			
Rod 21 use 0.58 22–23 · 4.00 m HEB 220 N -824.47 kN			
Rod 18 use 0.31 19–20 · 4.00 m HEB 220 N -419.88 kN			
Rod 29 use 0.31 30–31 · 4.00 m HEB 220 N -419.88 kN			

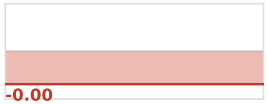
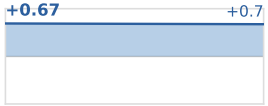
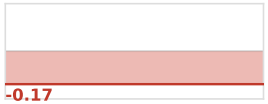
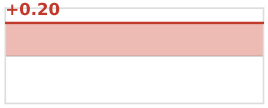
	Where	Shear V (kN)	Moment M (kN·m)
Rod 9 use 0.71 9–10 · 4.00 m HEA 220 N +899.18 kN			
Rod 6 use 0.71 6–7 · 4.00 m HEA 220 N +899.18 kN			
Rod 10 use 0.65 10–11 · 4.00 m HEA 220 N +824.23 kN			
Rod 5 use 0.65 5–6 · 4.00 m HEA 220 N +824.23 kN			
Rod 4 use 0.57 4–5 · 4.00 m HEA 220 N +719.38 kN			
Rod 11 use 0.57 11–12 · 4.00 m HEA 220 N +719.38 kN			
Rod 2 use 0.34 2–3 · 4.00 m HEA 220 N +419.68 kN			
Rod 13 use 0.34 13–14 · 4.00 m HEA 220 N +419.68 kN			

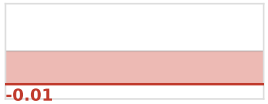
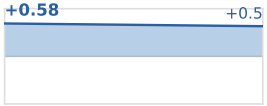
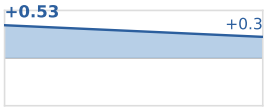
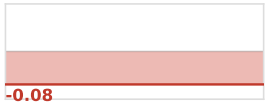
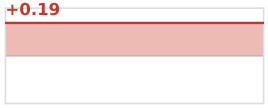
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 12 use 0.47 12–13 · 4.00 m HEA 220 N +584.55 kN			
Rod 3 use 0.47 3–4 · 4.00 m HEA 220 N +584.55 kN			
Rod 14 use 0.17 14–15 · 4.00 m HEA 220 N +225.18 kN			
Rod 1 use 0.17 1–2 · 4.00 m HEA 220 N +225.18 kN			
Rod 35 use 0.51 35–18 · 4.00 m SHS 120x120x5 N +223.25 kN			
Rod 62 use 0.51 48–32 · 4.00 m SHS 120x120x5 N +223.25 kN			
Rod 37 use 0.45 36–19 · 4.00 m SHS 120x120x5 N +195.15 kN			
Rod 60 use 0.45 47–31 · 4.00 m SHS 120x120x5 N +195.15 kN			

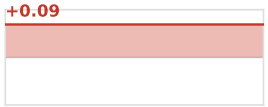
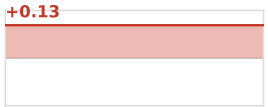
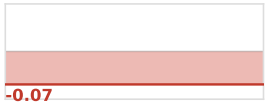
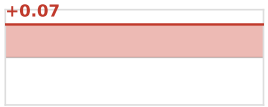
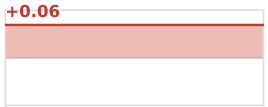
	Where	Shear V (kN)	Moment M (kN·m)
Rod 34 use 0.54 1–35 · 4.00 m SHS 120x120x5 N -165.26 kN			
Rod 61 use 0.54 15–48 · 4.00 m SHS 120x120x5 N -165.26 kN			
Rod 39 use 0.38 37–20 · 4.00 m SHS 120x120x5 N +164.41 kN			
Rod 58 use 0.38 46–30 · 4.00 m SHS 120x120x5 N +164.41 kN			
Rod 67 use 0.62 35–19 · 5.66 m SHS 150x150x6 N -274.65 kN			
Rod 93 use 0.62 48–31 · 5.66 m SHS 150x150x6 N -274.65 kN			
Rod 49 use 0.05 9–42 · 4.00 m SHS 120x120x5 N +13.88 kN			
Rod 46 use 0.05 7–41 · 4.00 m SHS 120x120x5 N +13.88 kN			

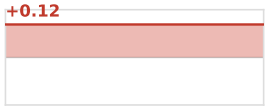
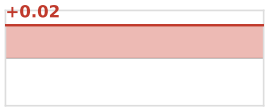
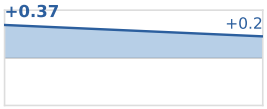
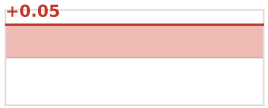
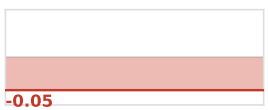
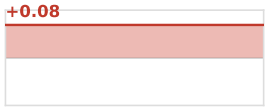
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 66 use 0.45 34–1 · 5.66 m SHS 150x150x6 N +316.03 kN			
Rod 96 use 0.45 49–15 · 5.66 m SHS 150x150x6 N +316.03 kN			
Rod 36 use 0.43 2–36 · 4.00 m SHS 120x120x5 N -134.18 kN			
Rod 59 use 0.43 14–47 · 4.00 m SHS 120x120x5 N -134.18 kN			
Rod 81 use 0.05 42–25 · 5.66 m SHS 150x150x6 N -22.84 kN			
Rod 79 use 0.05 41–25 · 5.66 m SHS 150x150x6 N -22.84 kN			
Rod 41 use 0.31 38–21 · 4.00 m SHS 120x120x5 N +134.67 kN			
Rod 56 use 0.31 45–29 · 4.00 m SHS 120x120x5 N +134.67 kN			

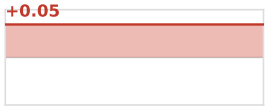
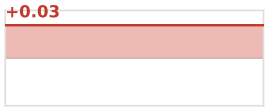
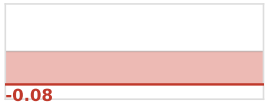
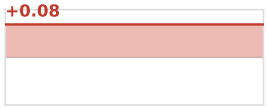
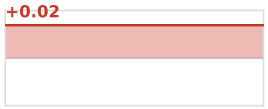
	Where	Shear V (kN)	Moment M (kN·m)
Rod 65 use 0.70 34–18 · 5.66 m SHS 150x150x6 N -316.46 kN			
Rod 95 use 0.70 49–32 · 5.66 m SHS 150x150x6 N -316.46 kN			
Rod 94 use 0.39 48–14 · 5.66 m SHS 150x150x6 N +274.86 kN			
Rod 68 use 0.39 35–2 · 5.66 m SHS 150x150x6 N +274.86 kN			
Rod 69 use 0.52 36–20 · 5.66 m SHS 150x150x6 N -232.71 kN			
Rod 91 use 0.52 47–30 · 5.66 m SHS 150x150x6 N -232.71 kN			
Rod 51 use 0.05 10–43 · 4.00 m SHS 120x120x5 N -15.06 kN			
Rod 44 use 0.05 6–40 · 4.00 m SHS 120x120x5 N -15.06 kN			

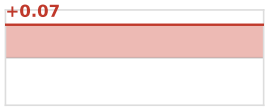
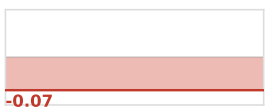
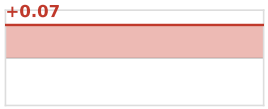
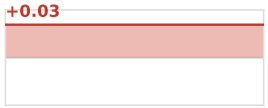
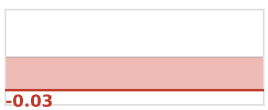
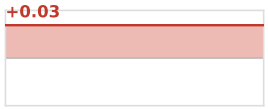
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	Where	Shear V (kN)	Moment M (kN·m)
Rod 71 use 0.42 37–21 · 5.66 m SHS 150x150x6 N -190.35 kN			
Rod 89 use 0.42 46–29 · 5.66 m SHS 150x150x6 N -190.35 kN			
Rod 42 use 0.13 5–39 · 4.00 m SHS 120x120x5 N -45.17 kN			
Rod 53 use 0.13 11–44 · 4.00 m SHS 120x120x5 N -45.17 kN			
Rod 38 use 0.33 3–37 · 4.00 m SHS 120x120x5 N -105.02 kN			
Rod 57 use 0.33 13–46 · 4.00 m SHS 120x120x5 N -105.02 kN			
Rod 84 use 0.09 43–9 · 5.66 m SHS 150x150x6 N +63.57 kN			
Rod 78 use 0.09 40–7 · 5.66 m SHS 150x150x6 N +63.57 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 43 use 0.23 39–22 · 4.00 m SHS 120x120x5 N +104.81 kN			
Rod 54 use 0.23 44–28 · 4.00 m SHS 120x120x5 N +104.81 kN			
Rod 70 use 0.33 36–3 · 5.66 m SHS 150x150x6 N +233.09 kN			
Rod 92 use 0.33 47–13 · 5.66 m SHS 150x150x6 N +233.09 kN			
Rod 83 use 0.14 43–26 · 5.66 m SHS 150x150x6 N -63.10 kN			
Rod 77 use 0.14 40–24 · 5.66 m SHS 150x150x6 N -63.10 kN			
Rod 75 use 0.24 39–23 · 5.66 m SHS 150x150x6 N -105.86 kN			
Rod 85 use 0.24 44–27 · 5.66 m SHS 150x150x6 N -105.86 kN			

One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 80 use 0.03 41–8 · 5.66 m SHS 150x150x6 N +23.32 kN			
Rod 82 use 0.03 42–8 · 5.66 m SHS 150x150x6 N +23.32 kN			
Rod 40 use 0.23 4–38 · 4.00 m SHS 120x120x5 N -74.98 kN			
Rod 55 use 0.23 12–45 · 4.00 m SHS 120x120x5 N -74.98 kN			
Rod 73 use 0.33 38–22 · 5.66 m SHS 150x150x6 N -148.05 kN			
Rod 87 use 0.33 45–28 · 5.66 m SHS 150x150x6 N -148.05 kN			
Rod 72 use 0.27 37–4 · 5.66 m SHS 150x150x6 N +190.74 kN			
Rod 90 use 0.27 46–12 · 5.66 m SHS 150x150x6 N +190.74 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 52 use 0.15 43–27 · 4.00 m SHS 120x120x5 N +74.52 kN			
Rod 45 use 0.15 40–23 · 4.00 m SHS 120x120x5 N +74.52 kN			
Rod 50 use 0.10 42–26 · 4.00 m SHS 120x120x5 N +46.36 kN			
Rod 47 use 0.10 41–24 · 4.00 m SHS 120x120x5 N +46.36 kN			
Rod 76 use 0.15 39–6 · 5.66 m SHS 150x150x6 N +106.31 kN			
Rod 86 use 0.15 44–10 · 5.66 m SHS 150x150x6 N +106.31 kN			
Rod 74 use 0.21 38–5 · 5.66 m SHS 150x150x6 N +148.48 kN			
Rod 88 use 0.21 45–11 · 5.66 m SHS 150x150x6 N +148.48 kN			



Rod 8

utilisation 0.75

H1-1a, axial + bending

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Rod 8
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Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 943,766 N / 6,156 mm²
      = 153.31 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.31 / 211.50
      = 0.725

 4. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 493,734 mm³
      = 104.425 kN·m

 5. Axial + bending  [H1-1a]
      Pr/Pc + 8/9 (Mry/Mc + Mrx/Mcw)
      = 0.725 + 8/9 (0.000/104.425 + 0.000/104.425)
      = 0.754

 6. Governs
      the largest ratio
      = H1-1a, axial + bending
      = 0.754

utilisation 0.754 -- OK
```



Rod 7

utilisation 0.75

H1-1a, axial + bending

```
Rod 7
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 943,766 N / 6,156 mm²
      = 153.31 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.31 / 211.50
      = 0.725

 4. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 493,734 mm³
      = 104.425 kN·m

 5. Axial + bending  [H1-1a]
      Pr/Pc + 8/9 (Mry/Mc + Mrx/Mcw)
      = 0.725 + 8/9 (0.000/104.425 + 0.000/104.425)
      = 0.754

 6. Governs
      the largest ratio
      = H1-1a, axial + bending
      = 0.754

utilisation 0.754 -- OK
```